

GJØVIK UNIVERSITY COLLEGE

BACHELOR THESIS

Snowman

A complete, effective and secure rule management system for Snort

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Preface

This is the preface.

Acknowledgements

We would like to thank our mentor from Gjøvik University College, prof. Slobodan Petrovic, for his help and input during the project. We would also like to thank BKI and especially Jarle Kittilsen for assignig us this project and the help with forming requirements and the software testing.

This project is a rule management system for SNORT®[1], henceforth referred to as Snort.

[1] Snort® is a registered trademark of Sourcefire, Inc.

Glossary

This chapter contains an alphabetical list of technical terms used throughout the report along with their definition.

CLI Command-line interface.

Daemon A computer program that runs as a background process, usually without any user interaction.

DOM Document object model.

HDD Hard disk drive.

HTML Hypertext markup language.

HTTP Hypertext transfer protocol.

I/O Input / Output, used to describe data flow to and from devices such as harddrives.

IDS Intrusion detection system.

MD5 A cryptographic hash function producing a 128-bit hash value.

MVC Model-view-controller.

Open Source Software Software where the source code is publicly available and is licensed in such a way that the copyright holder grants the rights to study, alter and redistribute the software to anyone for any purpose.

RAM Random access memory.

RPC Remote procedure call.

RPM Rounds per minute.

Sensor A computer system wherein Snort and/or other security software is recording network traffic and produces alerts when malicious content is detected.

SSD Solid state drive.

SSL Secure sockets layer.

URL Uniform resource locator.

XML Extensible markup language.

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1 INTRODUCTION

1.1 Document Structure

TBC: How have we organized this document.

1.1.1 Special styles used

Software names are written in a different font throughout the report, like so: Snowman. Source code is listed in an enclosed box with syntax highlighting (listing 1.1). Source code can also appear in the text with the same syntax highlighting as the listings, especially note classnames which will appear as so: `ClassName`.

```
1 def printHello():
2   # Print a message
3   print 'Hello World!'
```

Listing 1.1: Source code example.

1.2 Terminology

Edit: Moved to glossary.

1.3 Project Background

The Norwegian Armed Forces Cyber Defense (CYFOR) unit for Computer Network Defense (CND), Avdeling BKI, is responsible for detecting and stopping cyber-attacks against the critical infrastructure and command and control systems utilized by the Norwegian military. One of the tools used to carry out this task is the open source IDS Snort [1].

Snort is a leading open source IDS (ref?) that is used to detect potentially unwanted or malicious network traffic based on predefined patterns described in signatures. There are both free and commercial signatures available for Snort and as Snort has been highly adopted by the security community, these signatures are continually updated and maintained. Using Snort has been plagued for years by the fact that it does not have a good system for handling, updating or maintaining the signatures (henceforth called rules and rule sets) used for detection, thus making it cumbersome to utilize Snort for security analysts (ref?).

Avdeling BKI has therefore tasked a group of students from Gjøvik University College (GUC) to design and develop a more functional, effective and secure rule management system for Snort that can continue to live on beyond this project.

1.4 Previous Work

Research [2] has shown that a good rule management system for Snort will improve the workflow and efficiency of the IDS significantly, and a prototype for such a system was built in 2012 called Bring Home The Bacon [2] (henceforth referred to as BHTB). This prototype has been in use at Avdeling BKI since 2012, but is suffering from lacks in functionality and efficiency due to being a proof-of-concept prototype. BHTB has, however, proven that there is a significant gain from having an intuitive Graphical User

Interface (GUI) for managing rules. A GUI facilitates the work of the operator, as work is shifted from complex operations in a CLI to automatic execution of common routines triggered by buttons.

There was another similar project conducted at Gjøvik University College in 2012 called Snortmanager [3] that also deals with management of Snort rules. The Snortmanager project files are available as open source software[1], but no usage instructions are available. The source code has not been maintained since its initial release and is designed for a specific company in mind. Enabling, disabling and configuring rules is also cumbersome in this system as it is based on plaintext arguments in "policies" that users must construct themselves [3]. The rule distribution is also done manually from the files generated. This makes Snortmanager something not very usable for Avdeling BKI and their routines.

Several add-ons and other Snort-related software exist[2], and a few of these are built to manage Snort rules, most notably Pulled_Pork [3] and Oinkmaster [4]. The key differences between all software mentioned in this section and Snowman will be further explained in chapter 2.

1.5 Target Audence

The software this project has produced is targeted at users of the Snort IDS in a larger production environment with many sensors and networks. More specifically, that target audience usually consists of security experts with a high degree of knowledge of Snort and computers in general.

This report is targeted at those that may be interested in discovering just how complex the rule structure of Snort really is, and academics that are interested in reading about software performance and how complexity plays into it.

1.6 Project Objectives

1.6.1 Result Objectives

We reiterate from the project plan[link to project plan] that this project has a goal of researching, designing and developing a system that as a minimum:

- Is more efficient than BHTB.
- Is developed as an engine that responds to API calls.
- Is modular in design and allows future changes and additions to not affect existing functionality.
- Is scalable enough to handle both small and large sets of IDS sensors and rules in a production environment.
- Is able to download rules and rule sets from multiple customizable sources.
- Is able to distribute different rules to different IDS sensors.
- Can download and distribute rules automatically at set intervals.
- Is capable of exporting rules for sharing.

[1] http://github.com/stianja/SnortManager
[2] http://www.snort.org/snort-downloads/additional-downloads
[3] http://code.google.com/p/pulledpork
[4] http://oinkmaster.sourceforge.net

- Has the capability to activate, deactivate, filter and suppress rules on a per sensor basis, in a way that is user friendly.
- Supports multi-user interaction.
- Can efficiently and safely work with different installations of Snort and various other third party tools, such as SNORBY[4].
- Is easy to install.
- Has an internal logging system.
- Is well documented and can be easily adopted by anyone as open source software.

In addition, the project will attempt to add the following functionality if time permitting:

- A graphical user interface.
- Said user interface must be user friendly and intuitive, and require only limited knowledge of Snort rules from the user.
- The system can also distribute rules to Suricata[5], another Snort-like IDS, which can operate with the exact same rules as Snort.
- Implement security features that ensure confidentiality of rules.
- The project will investigate and potentially implement multi-threading technology or other optimization techniques to exploit modern hardware capacities.
- The system is capable of validating and checking rules for errors before they are distributed to sensors.
- The system supports commenting of the rules.
- The system supports writing custom rules and editing existing rules.

Once the system has been developed, it will be released as an Open Source Software to the public in a well-documented format.

1.6.2 Effect Objectives

Forming baselines from the current prototype BHTB and from Snortmanager, the project had a goal to further increase the efficiency of rule management by investigating and implementing programming techniques such as multi-threading and caching to reduce latency and delays in the system by as much as 50%.

As a very minimum the software efficiency must be linear or better, meaning if 1 action takes 1 second, 10 actions must take 10 or less seconds.

To measure our success in reaching these goals, we conducted a software efficiency test, comparing the three programs against each other using a range of test cases.

1.7 Project Description

In broad strokes, this project has been about designing a system that provides a simple and efficient way of handeling rule management for the Snort IDS. The project is the spiritual successor to BHTB. Essentially taking good ideas from BHTB, adding a few of our own, and creating a more adoptable open source rule management system for Snort.

We utilized the latest stable version of Python as our programming language. C++ was also considered, but due to the limited time at our disposal, we felt we could create

more functionality with Python, due to it requiring less memory management and the general “programmer friendly” ways of doing things.

The system required a database to function efficiently and the project primarily utilizes MySQL, as it is the de facto standard for the Linux platform. To make our database operations simpler, we’ve utilized a framework called Django[6] which offers us an abstraction layer on top of the database, so that we primarily only work with objects.

Furthermore, the project also consisted of creating a GUI that is user friendly and responsive.

The software that was created in this project, though still lacking some features and contains some minor bugs, largely fulfils the project goals outlined in section 1.6. We did not fulfil the goal of producing software that has reduced latency and delays, but we are confident that this is because we have sacrificed efficiency for functionality. We also feel that the functionality gained outweighs the efficiency lost and the efficiency curve is still very much within the linearity objective, as you can read about in chapter **??**.

Because the project has been so performance oriented, the report has evolved to also contain a small study into software performance. We’ve discovered that it is quite difficult to achieve performance in software when the expectations of functionality is high. We’ve also discovered the importance of conducting software performance testing before it is released, rather than after. Because we conducted a performance test on our software, we were able to implement optimizations that drastically improved the performance of one of our big operations within the software, the rule updating process.

1.8 Academic Background

This project has probably benefited greatly from the fact that all three project members are all taking different computer science courses. One member has studied computer engineering, the second has been learning the operations trade while the third has been studying information security. This has given us a unique opportunity to draw from each others strengths and approach the project from different perspectives. We have also benefitted from the diverse prior experience in the group, two of the members spent the year prior to this project at an internship at CERN in Switzerland. The third member of the team has worked for Avdeling BKI for the past 7 years prior to this project and has good knowledge of what the client needs and experience from the information security field in general.

1.9 Framework

1.9.1 Software Development Methodology

As this project has been very time restricted, the development needed be as efficient as possible. We therefore had to find a model that utilized the manpower we had available in the best possible way. The size of the group was quite small, so big and complex development models would add too much bureaucracy to coordinate the project and was thus overkill.

As we wanted to have a lightweight development model, some form of an agile[7] development model with a set of development cycles was preferred. This way, each group member could take an active part in the project and all members could feel a sense of ownership.

Developing iterative and incrementally, we got the best from both the agile devel-

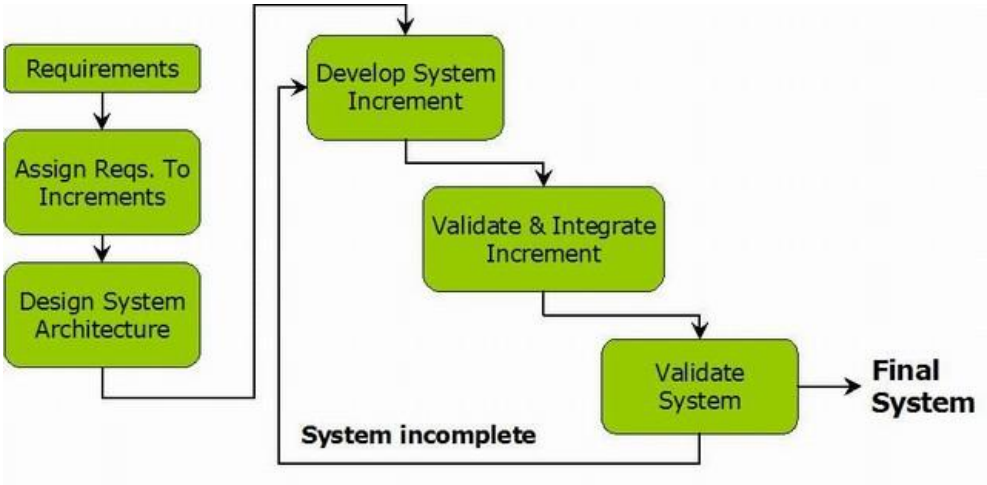


Figure 1: Development Model

opment world and from the very well defined waterfall approach[8]. We were able to get feedback throughout the project, and have a structured and reasonable way of actually doing changes to the plan while the project was running. At the same time we could be able to maintain the overall plan within cycles, so that it was possible to work individually, while still being well coordinated.

To be able to develop all the modules of the project within the timeframe, each development cycle consisted of multiple modules or functionality developed in parallel. This allowed us to maximize our utilization of time and personnel.

The start of each cycle consisted of a meeting where we defined the goals for that cycle and identifying the individual tasks that we needed to get done. This is similar to the start of a sprint in the development model Scrum[9].

In the development part of the cycle, we worked individually on the tasks that were defined in the planning. Regular meetings and cooperative programming within the group ensured that we all pulled the project in the same direction.

At the end of each cycle we would have a working set of software, which ideally complied with all the goals set in the start of each cycle. We held meetings with Avdeling BKI between each cycle to present the progress, and to let them influence the direction of the project.

After all the development cycles had been completed, we then made a final assurances that everything worked as intended and that all modules coexisted in harmony. This part of the development also consisted of some hands-on testing by the client and performance testing.

1.9.2 Schedule

We planned this project to have a fairly comprehensive initial phase for mapping and designing of the system. We spent the better part of the first two weeks figuring out how to implement the system. After the two weeks and before we started actually programming, we held a “project decision meeting" with the client to confirm that we had modelled the system correctly, to get feedback and to get the final go-ahead.

We then began the implementation phase which consisted of development cycles as

described above in section 1.9.1. This phase was devided into five parts, consisting of four development cycles and a final cleanup/testing cycle. All the development cycles lasted two weeks, aside from the last development cycle and the cleanup/testing cycle, which lasted one week each.

During those cycles we conducted meetings with both our mentor and our client every two weeks, timed so that these meetings took place on alternating weeks. This ensured that we got continual feedback and advice throughout the entire process.

After these weeks, we completely shifted focus from the software to the project report and spent the final 5-6 weeks working on that. Because we had done a thorough job during the first few weeks of the project, we had a lot of the contents for the report more or less created.

1.9.3 Project Organization

We organized the project work a little differently throughout the project, reflecting on the fact that we had multiple phases. During the first few weeks we spent most of the time together in a group room, working together and discussing things. Once most things had been worked out and the implementation phase began, there was more room and need to work individually to produce code. We still gathered together every so often to syncronize our progress and plan the next course of action.

Throughout the project, most decisions have been met either by confering with the client or achieving consensus within the group.

2 EXISTING SOLUTIONS

To identify exactly what Snowman does differently than other, similar software, we will quickly go over their functionality and how they differ.

2.1 Functionality Comparison Chart

To easily visualize how the various software is different from each other, we have gathered key functionality from all of them and created a list of support for them.

***putt inn tabell her ***

2.2 Bring Home The Bacon

Bring Home The Bacon was a byproduct prototype of the Master thesis written by Henriksen in 2012[2]. In the thesis, Henriksen sought to prove that having a central, graphical-oriented, system to administer multiple Snort sensors and their rules results in a signficant improvement in user workflow over the more traditional command line environment.

Henriksens tests showed that such a system could double the users work efficiency, if not more, as the command line environment would scale very badly once the number of sensors increases. Henriksen also outlines a rule management process, which we have taken into account in our initial work to map the program workflow in section 3.1.1.

Despite being a prototype, BHTB does offer the essential functionality required for managing rulesets on multiple sensors. It can display a list of rules where you can turn those on or off on different sensors. It can also display a list of rulesets and you can turn those on or off, but that only propagates down to the rules, turning those on or off. That has been a drawback for BHTB, as you loose whatever preferred setup you had if you simply want to turn one ruleset on or off for a short while, and thus have to remember how it was set up before.

BHTB also lets you add thresholds and suppressions, but the system does not do any validation of the input, thus leaving the Snort sensor vulnerable to crashes[1] if the user inputs something wrong.

On the technical side, BHTB is programmed with Python and uses the framework Tornado[10] as its webserver, which works similar to Django with it's URL API system which you can read about in section **??**. It also uses SQLite[11] as it's database, which has limited abilities when it comes to relations and more complex data structures.

BHTB really begins to show it's limitations once the system begins to be populated by a lot of rules[2] and many sensors, thus having a scalability problem. Avdeling BKI has used BHTB in a production environment since 2012 and experiences the software as slow and cumbersome.

2.3 Snortmanager

Beskriv hva den gjør

[1] Snort will simply not restart if something in the configuration files is malformed.
[2] In the five digit range.

2.4 Other Programs of Note

There are two other pieces of software that also should be mentioned. Pulled_Pork and Oinkmaster are both programs that can download rules and update your Snort sensor, and even do some of the more advanced features you will find in Snowman and BHTB. Pulled_Pork even has support for the Shared Object rules[12], something this project has not taken into account due to complexity.

However, both of these programs suffer from the fact that they are command line oriented programs. They are both very good at what they do, but do not have the userability and efficiency that you get from a GUI with buttons. In a limited environment with one or very few Snort sensors, they are probably a better choice for rule management than something like BHTB or Snowman. But once the environment grows larger, with tens or even hundreds of sensors, they would start to slow the user down.

2.5 Research on Software Performance

Since this project also investigates software performance, it is relevant to take a look at what others in the field write on the subject. We have noticed that most development reports from GUC do not really mention the subject, probably due to most student projects being more or less agile[7] development projects. This is something mentioned in [13], that agile programming is becoming mainstream and by it's nature is a concern for the future of software performance engineering, as the primary focus of an agile project is to pump out functionality.

Early on in the project we noted that Snowman is very I/O bound, as it does a lot of database operations. So much so that there is a very noticable difference in performance between traditional rotational harddrives and SSD, as seen in section **??**.

Myers speaks of this in his Master thesis[14]. His tests show that SSD has a significant advantage over HDD when it comes to random read performance but not when it comes to write performance. This is to be expected as his tests were conducted in 2008 and SSD technology was fairly new back then, suffering from their different way of writing data[3]. As operating systems, drivers and disk controllers has improved, SSD's are now superior to HDD's in most tasks.

When we identified software performance testing as one of the goals of our project, we needed to identify a good testing method. Sadly, there is not a whole lot written on the subject, assuming most testing just follows the scientific method or common sense. However, the articles [15] and [16] gave us a few pointers towards developing the test cases described in chapter **??**. Both articles also stress the importance of performance testing during the development rather than after it, as it gives you time to actually implement optimizations and improvements.

[3] Features such as TRIM was introduced shortly after Myers thesis.

Feature	Snowman	BHTB	Snortmanager	Pulled_Pork	Oinkmaster
Can download rules from source	✓	✓	✓	✓	✓
Can store rules	✓	✓	✓	✓	✓
Can register and administer sensors	✓	✓	✓		
Has a graphical user interface	✓	✓	✓		
Can automatically update database	✓	✓		✓	✓
Can automatically syncronize sensors	✓	✓		✓	✓
Can turn on or off rules	✓	✓			✓
Works with multiple sensors	✓	✓			✓
Can syncronize rules to sensors	✓	✓			✓
Displays change history of updates	✓	✓			✓
Can store meta data about rules	✓	✓			
Can turn on or off whole rulesets	✓	✓			
Can turn on or off rulesets on individual sensors	✓	✓			
Can add filters and suppressions to rules	✓	✓			
Can add filters and suppressions to rules on a per sensor basis	✓	✓			
Has an internal logging system	✓	✓			
Can display list of rules in the system	✓	✓			
Can display meta data in the system	✓	✓			
Normalized database	✓		✓	N/A	N/A
Does not require root to interact with Snort.	✓				
Turning rulesets on or off is non-propagating	✓		N/A	N/A	N/A
Can import filters and suppression from rule files	✓		✓	✓	✓
Has input validation	✓			N/A	N/A
Supports multiple database technologies	✓			N/A	N/A
Handles multi line rules	✓				✓
Can skip certain files	✓				✓
User can approve changes before sensors are updated	✓				✓
Is easy to install	✓				
Can import rules from arbitrary sources	✓ 9				
Can organize rulesets in a hirarchy	✓				
Can create new custom rulesets	✓				

3 REQUIREMENTS SPECIFICATION

Many of the requirements for this system stems from functions that already reside in BHTB, which our client wants either reproduced or improved upon. That means the requirments for this project is closely tied to the way BHTB operates and its general workflow. In this coming section we will outline the specifications we've gathered from BHTB and from conversations with the client.

3.1 Functional Requirements

3.1.1 Program Workflow

To develop our requirements, we decided to chart the program workflow in order to better understand what the system is expected to do. Since this was to be a highly user-oriented system we accomplished this by going over the old system, BHTB, step by step with the client and detailing the functions and various needs for improvement or additional functionality. From this, we developed a workflow chart that reveals the major system parts. This is in addition to taking into account the management process outlined in Henriksens thesis[2].

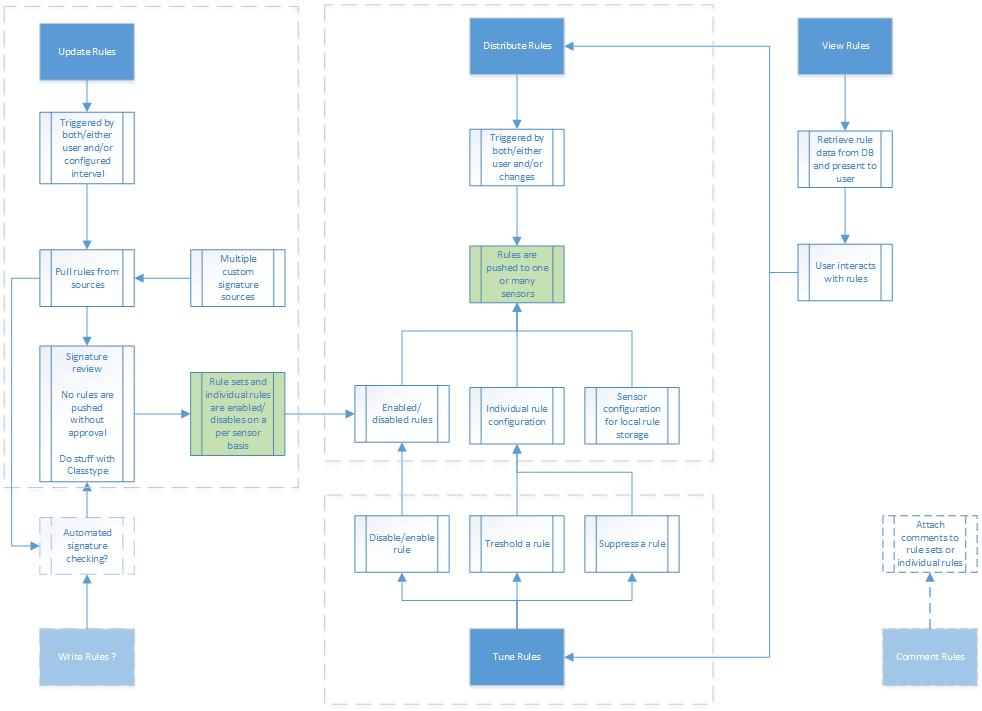


Figure 2: Program Workflow

To explain the workflow chart in figure 2, the blue boxes are the major system functionalities, the green boxes are the major goals of the system and the arrows show the direction of flow the system should have.

As you can see, if you trace the arrows, all of the program flow ends up with rules being syncronized to the sensors. This, of course, makes sense since that is the whole purpose of the system to begin with.

But the road to Rome consisted of a lot more functionality that needed to be in place, and that interacts with the rules, before they are ready to be handed over to Snort.

It's important to note that very early on, we decided that writing rules, automatically check them for errors and commenting them were nice-to-have (but still important) features that we could implement if we had time to do so. Hence they are greyed out in the drawings, and as you will notice further on in the report, was never developed due to time contraints.

From this workflow we could outline a rough modular design (figure 3) based on the naturally separate functions of updating rules, tuning them, distributing them and show them. We also added a database module, since most of our program was going to be centered around it.

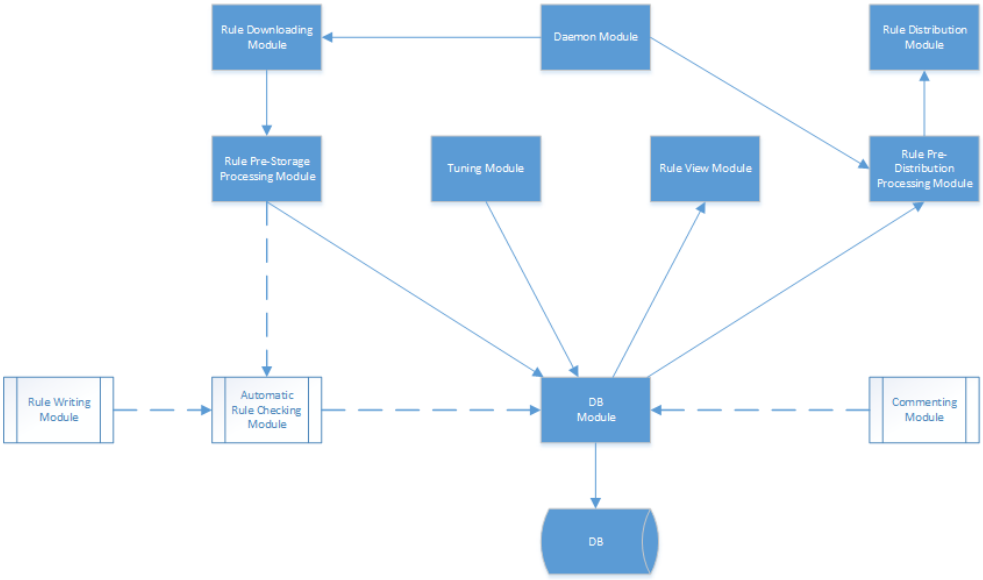


Figure 3: Program Modules

Rule Downloading Module

This module will handle all functionality related to downloading rules, including comparing MD5 checksums.

Rule Pre-Storage Processing Module

This module will essentially take downloaded rules, and parse and process them into objects that can be fed into the database.

DB Module

This module will handle all input and output to/from the database.

Rule Pre-Distribution Processing Module

This module will essentially take rules stored in the database and prepare them for distribution to sensors.

Rule Distribution Module

This module will be in charge of distributing rules out to the Snort sensors and handing the rules over to Snort itself.

Tuning Module

This module will handle all functionality related to turning rules on or off, filtering them or suppressing them.

Rule View Module

This module will contain all the functionality needed to display the system status and system functions to the user. It will be the primary way the user interacts with the system. Basically the GUI module.

Daemon Module

This module will run continually on the system to ensure that the system is responsive and to carry out any interval based tasks, such as automatic updates or distributions.

Commenting Module

This module will allow users to attach comments to rules.

Rule Writing Module

This module will handle functionality for writing custom rules and editing existing rules.

Automatic Rule Checking Module

This module will be responsible for verifying rules to ensure that there are no malformed or misconfigured rules that may potentially break or cause a halt on a Snort sensor.

3.1.2 Use Case Diagram

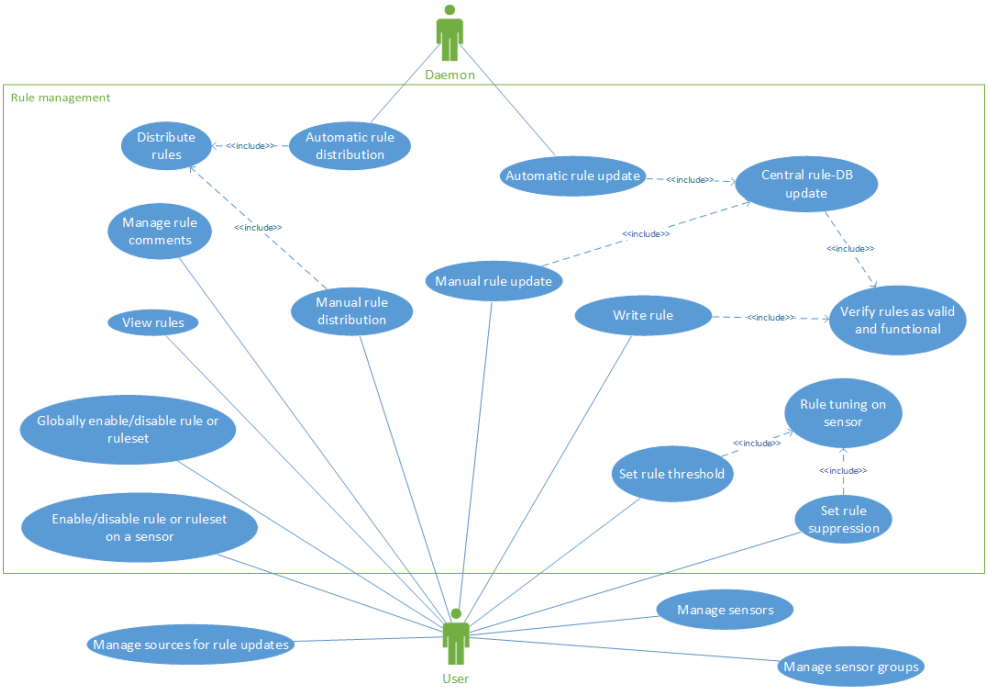


Figure 4: Use Case Diagram

3.1.3 High Level Use Cases

Looking at the arrows in the program workflow in figure 2, it is pretty apparent that the program workflow is heavily centered around the database module, and the major parts of the system revolves around the importing of rules and then distributing them. We therefore began developing our use cases around those major movements.

We've added the most central use cases for this system below, but additional use cases is available in appendix **??**.

Use case	**Manual rule update**
Primary actor	User
Purpose	Update the central rule database with the newest rules and revisions.
Description	The user commands the server to update the rules, either from a local file or from external sources. When the latter is the case, the server pulls rule files from all the sources specified in the list of sources. In both cases rules are processed and stored in the database.
Preconditions	Either a file or one or more sources must exist.
Postconditions	The central rule database is up to date with the newest rules and revisions.

Use case	**Automatic rule update**
Primary actor	Daemon
Purpose	Update the central rule database with the newest rules and revisions.
Description	At configured time-intervals, the system daemon triggers a rule database update. The server pulls rule files from all the sources specified in the list of sources to be updated at the given interval, processes the rules and then stores them in the database.

Preconditions	One or more sources must exist.
Postconditions	The central rule database is up to date with the newest rules and revisions

Use case	**Manual rule distribution**
Primary actor	User
Purpose	Distribute rules to sensors as per the settings in the database.
Description	The user sends a command to distribute rules to either all or specified sensors. The server then begins the distribute routine, and eventually all sensors will be up to date with the rules and rulesets they should have.
Preconditions	Rules and sensors must exist
Postconditions	Specified sensors contain rules as per settings in the central database.

Use case	**Automatic rule distribution**
Primary actor	Daemon
Purpose	Distribute rules to sensors as per the settings in the database.
Description	Configured at intervals, the system triggers a command to distribute rules to specified sensors. The server then begins the distribute routine, and eventually all sensors will be up to date with the rules and rulesets they should have.
Preconditions	Rules and sensors must exist

Postconditions	Specified sensors contain rules as per settings in the central database.

Use case	**Enable/Disable a rule or ruleset**
Primary actor	User
Purpose	Enable or disable one or more rule or ruleset on one or more sensors.
Description	The user either enables or disables a rule or a ruleset so that the rule is either on or off on certain sensors. This status is then shared with the sensor and the sensor syncronizes as needed.
Preconditions	Rules and/or rulesets must exist.
Postconditions	A rule or ruleset has been modified to be enabled or disabled and the change is syncronized to affected sensors.

Use case	**Set rule threshold or suppression**
Primary actor	User
Purpose	Set or remove a threshold or suppression for a rule on one or more sensors.
Description	User sends a command with either thresholding/suppression parameters or removal of a threshold or suppression for a rule on one or more sensors. The system updates the database and distributes the change to the sensor(s).
Preconditions	Rules and sensor(s) must exist.
Postconditions	Rule tuning is saved in the central database and distributed to affected sensors.

Use case	**View rules**
Primary actor	User
Purpose	Display rules to the user.
Description	Rules are retrieved from the database, processed for GUI format and then delivered to the GUI.
Preconditions	Rules must exist.
Postconditions	The user is informed of the current status of the central database and the status of rules and rulesets enabled/disabled on sensors.

3.1.4 Detailed Use Cases

We also worked on some detailed use cases on our more complex system modules. Again, more use cases are available in appendix B.

Use case	**Central rule-database update**
Primary actor	Server
Purpose	Update the central rule database with the newest rules and revisions.
Description	An update can be triggered either by the user or the daemon. The system retrieves the files in question, checks what is new and what is not and then updates the database accordingly.
Preconditions	Depending on the request, this process requires a local file or the server must be configured with external source(s).
Postconditions	The central rule-database is up to date with the latest rules and revisions.
Triggers	Manual rule update, Automatic rule update, Write rule

Basic Flow:

1. Update is triggered.
2. Rule Downloading Module compares hash values from external sources with last update.
3. Updated rule files are downloaded from sources. The files are validated and passed to the Pre-Storage Processing Module.
4. The new files are parsed for rule SID and revision number.
5. A list of rule SID and revision numbers are gathered from the database and these are compared to identify which rules are new or modified.
6. New or changed rules are then parsed into objects and these objects are passed to the Database/Core Module.
7. The Database/Core Module stores the new or changed rule objects into the database. The user is notified that the update is complete and given a list of changes.

Extensions:

1. 1. Update is triggered by user.
   2. Update is triggered by daemon.
3. 1. The hash-values from all sources are identical with previously stored values. Database is up to date and the update process terminates. User is notified that database is up to date.
   2. The hash value from one or more sources is newer than previously stored values. Database must be updated, and the user is notified that the database is not up to date and an update process has begun.

Error handling:

1. The source and database checksum is different but the source rules are older than what is stored in the database: *The program can still continue, because if the SIDs already exist they will not be added again and if the revisions are older the rules will also not be updated.*
2. A ruleset has changed its name or one or more rules are moved to a different ruleset: *Based on a configuration file, the affected rules will either be a) automatically moved to the new ruleset or b) stay put in their original ruleset. The user will be notified and can choose to reverse or distribute the changes in either case.*

Use case	**Distribute rules**
Primary actor	Server
Purpose	Distribute rules from the central database to the sensors.
Description	After a command to synchronise is sent either by the user or the daemon, the server calculates and organizes the central database data and synchronizes this information with the affected sensors.
Preconditions	One or more rulesets or rules must be present in the central database. One or more sensors must be registered.
Postconditions	The rules on the sensor(s) are synchronized to reflect those in the central database that are enabled/disabled for those sensors.
Triggers	Manual rule distribution, Automatic rule distribution

Basic flow

1. Synchronize signal is sent.
2. The system calculates and organizes the data for each sensor affected by the synchronization.
3. The system initializes a synchronization protocol with the affected sensor or sensors, sending a list of what the sensor should have.
4. The sensor calculates diff between should have and has.
5. Compare diff.
6. Send update.
7. The sensor includes the new changes into its database and files and tells the central system that it is now up to date.
8. The system carries out maintenance to reflect the synchronization.
9. The user is notified that all sensors affected are now up to date.

Extensions

1. 1. User sends a command to synchronize one, more or all sensors
   2. The system reaches an interval for synchronizing all sensors.
   enumi4
3. 1. The sensor content is different from what it should have, asks central system to send what it needs.
   2. The sensor content is not different from what it should have, no further action is required. This status is sent to the central system.
3. 1. The system retrieves and sends what the sensor asks for.
   2. The system terminates the synchronization protocol with the sensor in question.

Error handling

1. The change may never be synced to the sensor due to errors: *There must be an error protocol that is followed in this scenario.*

3.1.5 Domain Model

After outlining all the things above, we put together the concept model in figure 5 of the objects we would need in the system. Even though a Snort rule is quite simple in and of itself, there are a lot of meta information surrounding them, like rule revisions and rule classes. Thus our system needed at least 16 different classes in order to fullfil the basics of the required functionality.

3.2 Supplemental Requirements

In addition to the features discovered in the use cases, we added had some requirements we felt were needed, that can be broken down into a RUP FURPS+ [17] fashion.

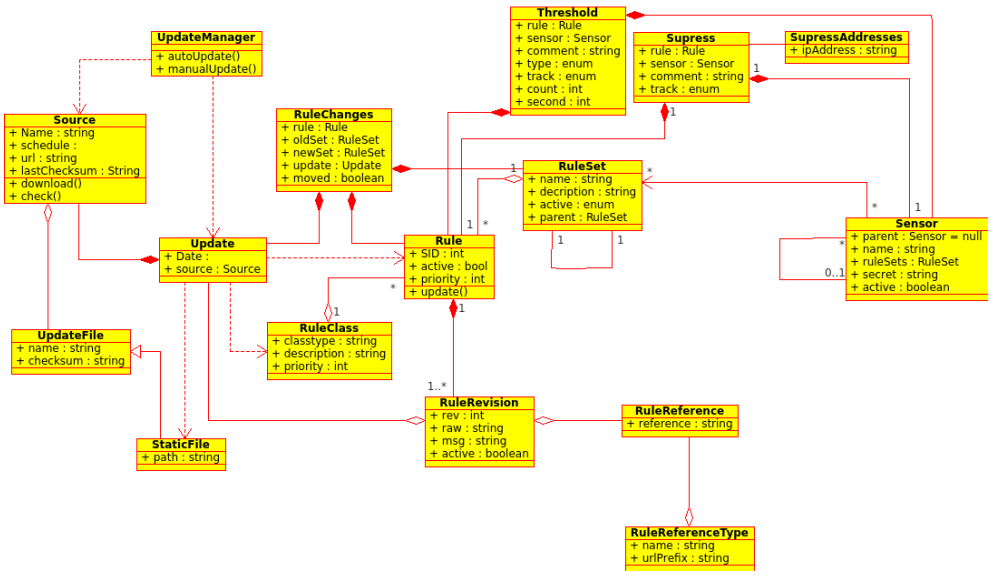


Figure 5: Conceptual Class Diagram

3.2.1 Functionality

- The system should be able to restart a Snort sensor in the event that one crashes.
- The system should be able to detect if a Snort sensor is not responsive over the network.
- The system must be portable to the majority of the popular linux distributions.

3.2.2 Usability

- The graphical user interface must be simple and clean, but still contain all the features any user might require to interact with the back-end system.
- The system interfaces, herein API and function calls, must be self-explanatory.

3.2.3 Reliability

- The system must not cause a SNORT sensor to crash.
- The system must be responsive to API calls at all times. In the event the system capacity limit is reached, new requests should be queued.

3.2.4 Performance

- The system should handle administration of at least 200 sensors.
- The database should be able to maintain at least 100 000 individual rules.
- Any operation in the system in normal operations (i.e. after the first big rule update) should take no longer than 1 second on average.
- The relationship between time and workload should be linear, meaning that if a single operation that takes 1ms, ten operations should take 10ms.

3.2.5 Security

- The link between the system server and the remote sensor must be secured in such a way that a man in the middle cannot read the transmission in plaintext.
- The remote sensor and the system server must verify that they are trusted parties in the system, through authentication.
- Certain important API calls must be protected against misuse from a non-trusted party.
- All processes and files related to them system must exert the principle of least privilege.

3.2.6 Interoperability

The system must be able to work with Snort in such a way that it can import new files to Snorts configuration files structure and must be able to reset Snort in such a way that the new configurations are loaded.

3.2.7 Licensing

The system will be released to the public as Open Source under the GNU General Public License v3 license[18].

3.3 Constraints

3.3.1 Platform

As mentioned in the Project Plan, Snort is primarily used in conjunction with the Linux/UNIX operating systems. For that reason, the project was restricted to only develop for that platform. More specifically, we limited ourselves to the Ubuntu and Debian operating system, which is the primary platform the client uses as well. We have, however, stived to make the software portable.

3.3.2 Data security

While the system itself will not handle any data that may comprise of personal data, it may contain rules and rule sets that are considered classified or confidential. The system has therefore been built with this in mind and ensures confidentiality within the boundaries of the system. The system does not guarantee the confidentiality of the system outside these boundaries.

These boundaries are defined as the program and its processes and threads, and the network link established between the system and remote sensors. The system assumes that the users have already established database security.

3.3.3 Graphical User Interface

The project has had its main focus on the systems functionality and efficiency, and developing a graphical user interface (GUI) has been considered a secondary task and was prioritized as such. None the less, a GUI for demonstrational purposes has been included.

3.4 Security Assessment

During the inital phase we conducted a small security assessment in order to be aware of and possibly fix any security issues the software might have. We did a small brainstorming

session and created a simple risk analysis table of the potential weeknesses we could find.

Threat	Component	Consequence	Probability	Risk	Strategy
Unauthorized users change data through the API/Web-interface	API/Web	8	5	6,5	Authentication/authoriz Django support for use and authentication.
An unauthorized user gains direct access to the database and is able to alter the data.	Central database	10	1	5,5	Assume database securi is in place.
Continually sending requests to the server, filling it up with requests to the point of causing a denial of service.	RPC Server	7	4	5,5	Authenticate requeste Rate-limiting on request logging.
An attacker inherits escalated privileges through exploiting the software if it is run as root or other high privilege user.	All processes	8	3	5,5	Follow least privileg principle. Python als protects against buff overflow
The SID/Rev of a malicious rule is newer than an excisting rule, with the purpose of overwriting the excisting rule.	Update module	7	2	4,5	Add possibility for th user to see changes befo they are commited.
In a scenario where the database is not on the localhost, a man-in-the-middle attack is possible.	Central database	6	3	4,5	Assume database securi is in place, encrypt tran missions.
Continually sending requests to the sensor client to update itself, filling it up with requests to the point of causing a denial of service.	RPC Client	5	4	4,5	Limit the allowed upda frequency, logging.
Some unauthorized machine poses as a legitimate sensor and gains access to rules through update requests.	RPC Server	5	2	3,5	Authenticate the sensors
During an update, one or more rules contains errors that causes Snort running on a sensor to crash or shut down.	Update module	8	4	6	Verify the rules.
A rule file is specially crafted to contain an SQL injection.	Update module	10	2	6	Use prepared statemen and never let input d rectly touch the databas
SQL injections, XSS og CSRF (cross site request forgery)	API/Web	5	5	5	Django has support fo protection against thes attacks. Validate input.

Table 11: Security Assessment

4 CONCEPTS OF SNORT

This chapter will give a brief introduction to the basic concepts of Snort.

4.1 Rules

A rule is basically a set of options that define ways of detecting, classifying and reporting a potential attack or threat on the network[1]. There are several types of options, but only the general rule options are relevant for this project. The general rule options provide information about the rule [12] (i.e. revision number, alert message). Other types of options, such as payload options which are actively used in detection, are not relevant for the management of rules and are thus not considered in this project.

A rule can either be enabled or disabled in the detection system, meaning that the rule will only have an effect on detection when it is enabled. Since this project is dealing with multiple sensors connected to one central database, this functionality has been expanded so that users can keep a rule enabled on one sensor and disabled on another. In addition, the user can also set an enabled/disabled flag on *rule sets*.

4.2 Rule sets

The term rule set, sometimes ruleset, is not formally defined in the Snort Users Manual, however it is used a few times to refer to the complete collection of official Snort rules[2] and to the complete set of rules applied on a Snort system. In practice, rules within a rule set are categorized by intrusion type (e.g. malware, trojan and worm). These groups of rules are often named by their intrusion type, and are also referred to as rule sets. Snowman and this report use only the latter definition of rule sets.

4.3 Generators

Generator is a term used for any subsystem of Snortwhich generates alerts from rules [12]. Each generator is identified by a unique generator ID. The rules subsystem, which has generator id 1, is the only relevant generator for this project.

4.4 References

Snort contains a plugin which allows rules to contain references to external attack identification systems which provide additional information about the alerts [12]. A reference is basically a URL string pointing to a webpage containing this information.

4.5 Event and detection filters

Filtering can be applied to rules in order to control the amount and frequency of alerts. Two types of filters can be applied to a rule: event filters and detection filters. A detection filter can only apear inline, whereas an event filter can only appear as a standalone

[1] A rule actually consists of a header and options [12]

[2] Also referred to as the Sourcefire VRT Certified Rules, see http://www.snort.org/snort-rules/

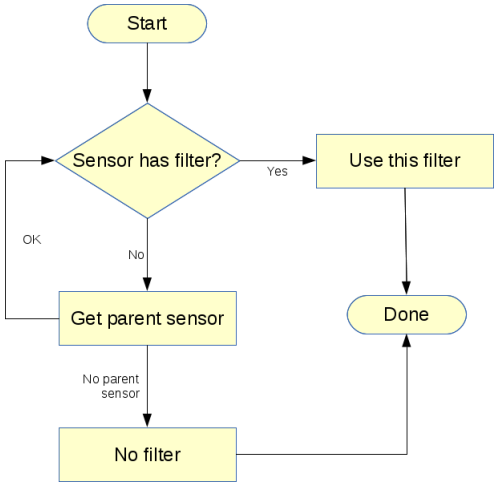


Figure 6: Flow

option[3].

All filters operate with a *count* and *seconds* variable, representing the number of events in a time interval and the duration of the time interval, respectively. Event filters are further divided into three types (table 12).

[3]An event filter can currently be written inline (called a rule threshold), but according to the Snort Users Manual this feature is deprecated and will not be supported in future releases.

Filter type	Alerts
Event filter	
Limit	One for every event when number of events $\leq$ C in interval S
Threshold	One for every C events in interval S
Both	One per interval S when number of events $=$ C
Detection filter	One for every event when *count* $>$ C in interval S

Note: C and S represent the count *and* seconds *values in the filter.*

Table 12: Effects of different filter types.

5 DESIGN

This chapter will present the design specification of Snowman. Snowman is developed as a distributed system (why?), with a central server which is responsible for the user-interface and the central core functions such as distributing rules to the sensors and downaload updates from external sources. A Snowman client, responsible for the communication with the central server, is installed on each sensor. The principal tasks of the client is to generate Snort rule files and respond to the server's status requests.

5.1 Server

The server is the central component of Snowman and generally consists of:

- Central rule storage
- Web-interface for user interaction
- Update-module to get updates from rule-sources
- RPC-interface to communicate with the clients

The central storage of rules would be a normal relation database. As we have decided to use Django as our framework, we have a great flexibility to which database we actually use. In the current version of Django (1.6) we might use MySQL, PostgreSQL, SQLite or Oracle.

For the RPC-Interface, we have decided to use XML-RPC. This is a very simple protocol using XML over HTTP to transfer requests and data to and from the server. To be able to keep an aspect of privacy, we encrypt the raw HTTP stream using SSL.

5.1.1 Architecture

The server architecture follows the layered approach of MVC (figure 7).

The web-interface uses pure Django logic. The HTML visible in the users browser is generated by the Django template-engine, which recieves its data from the views created by using ordinary pyhton code.

The XML-RPC server uses a handler which is administering the connections from the sensors, and exposing the relevant methods to allow the sensors do a synchronization.

The update is unpacking and preparing the update-files for parsing. It then handles the files to the parser. The parser is then handling the different parametres to the Updater, which in turn saves them to the database in an efficient manner.

Finally, common for all the interfaces is the Django Models. We use the database-abstractionlayer from Django for mapping the lines of a relational-database to django objects. This is the part that lets us write code once, and still be able to use several different database-engines.

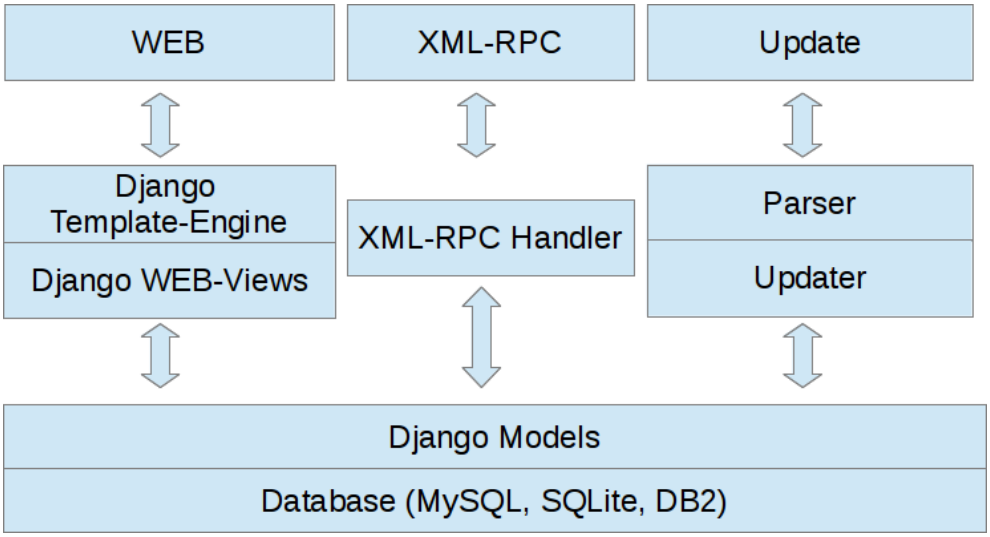


Figure 7: Server Architecture

5.1.2 Database

As the database of the server is quite complex, we have divided the drawings, so that we have one drawing per module.

Core

The core database-architecture used for storing the rules is as following:

The most central part of our system is the representation of a Rule. A single Rule might have several revisions, as we might want to keep some history when they get updated. Each revision have its own set of references attached.

A single rule also have a lot of other properties which it might share with other rules. A generator, a classification and a ruleset are examples of such.

Finally, the rules are placed onto a sensor by attaching ruleSets to the sensors.

Tuning

In addition to storing the raw rules, we would like to be able to store tuning-parametres on the rules. As sensors might have different tuning-parametres, we cannot add the tuning directly to the rules. We rather add the tuning to its own module:

Both filters, and the suppresses is connected to a Rule and a Sensor. This way we let the user spesify filters for the individual sensors if that is desired. If a sensor does not have a filter applied, but any parent of the sensor have one, the closest parent's filter will be applied.

Update

When rules are inserted to the database, we would like to have some traceability to be able to determine from where a rule have entered our system. To be able to do this, we have the update-module:

The structure of the database to update would let us store the update-sources, and add a update-object for each update ran from a given source. These update objects can in turn have a relation to rules, rulesets and rulerevisions, so that we can track what changes for each update.

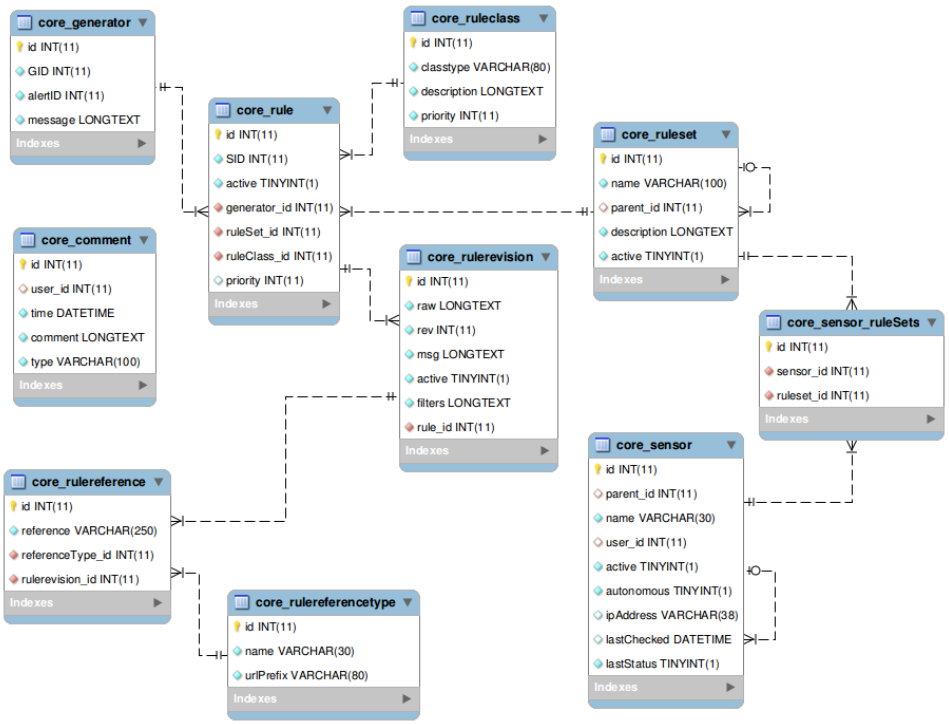


Figure 8: Core server-database

5.2 Client

As the client is just going to take care of getting rules for a single sensor from the central server, and should just apply these to one sensor, the architecture of the client can be much simpler than the one of the server.

5.2.1 Architecture

The MVC schema for the client would look like this:

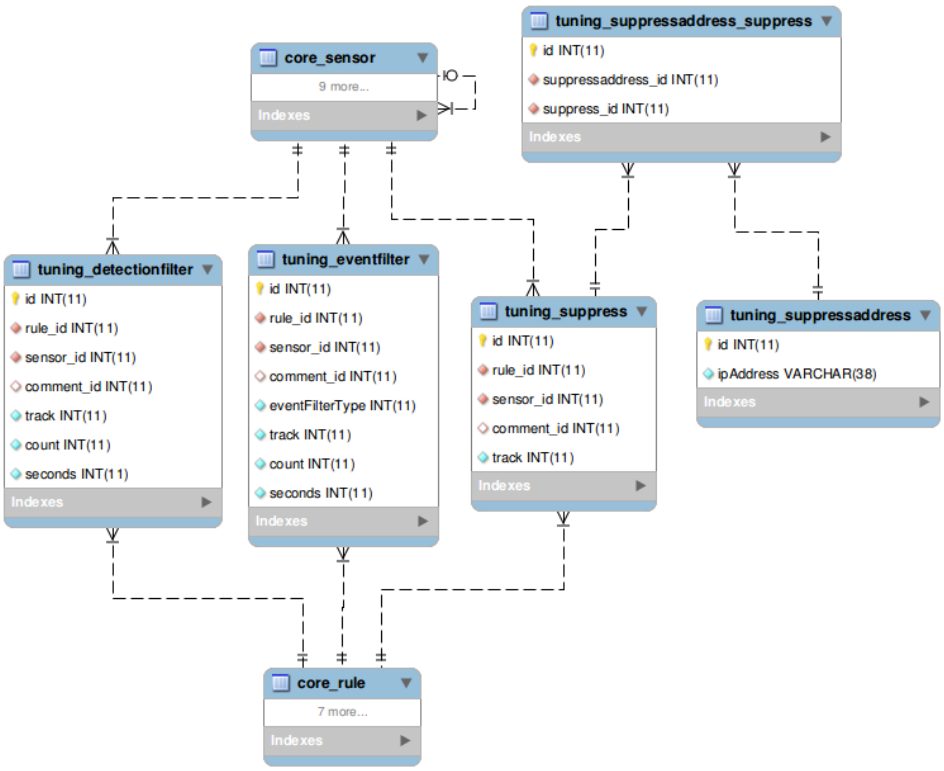


Figure 9: Tuning server-database

./img/Client-MVC.png

Figure 11: Client Architecture

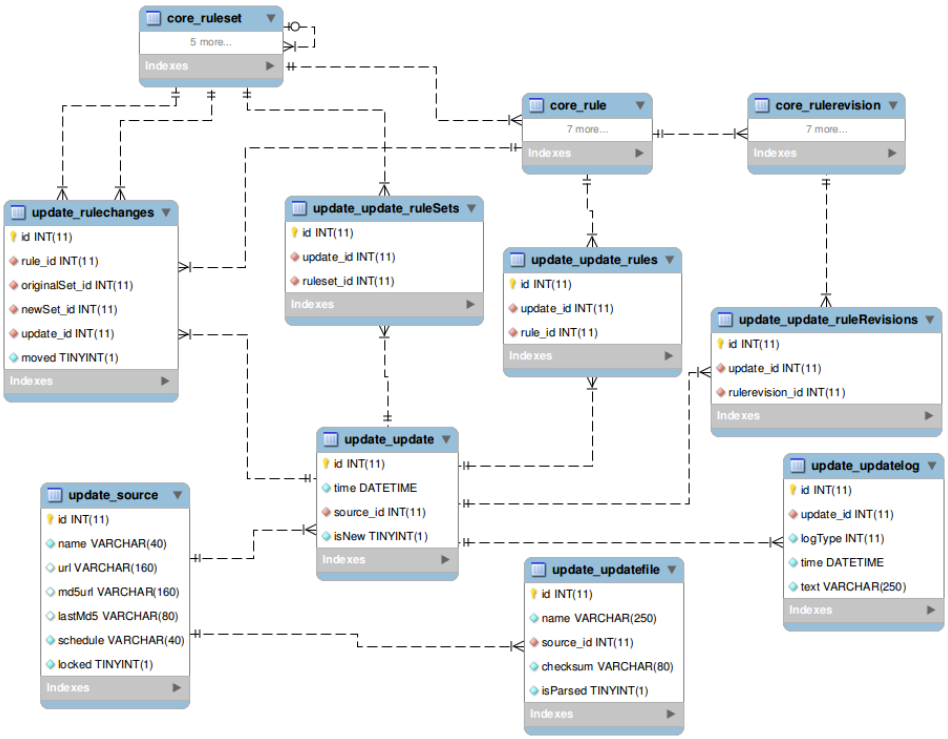


Figure 10: Update server-database

The update-handler of the client have both a XML-RPC Client and a XML-RPC Server. The client is used to connect to the central server, to do a rule-synchronization. The rpc-server is here so that the central server can connect to it and check that the client is alive, or to order an update of the rule-sets.

As the client should be as light-weight as possible, and the only thing we would like to have a framework for is the database-connection, we have decided to not pull inn Django for the client. Instead we use the database ORM for python called "SQL-Alchemy".

5.2.2 Database

The database on the client would be significantly simpler than the one on the server, as we only care about the rules which should be present on a given sensor.

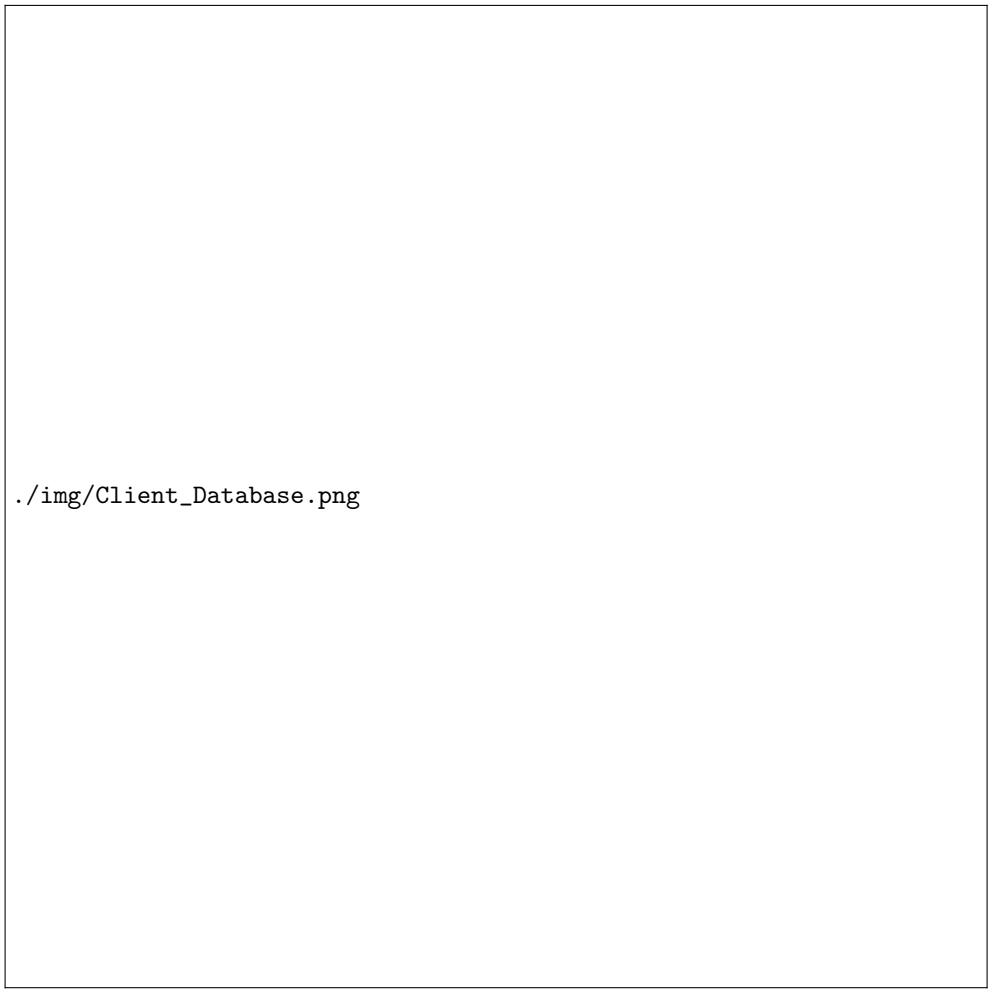


Figure 12: Client Database

The central part of the datastructure is the rule table. This table contains all the parametres directly related to the rule. In addition we have seperate tables to all the parametres modifying the rule (like filters), or that is common between several rules (like rulesets).

6 IMPLEMENTATION

This chapter will focus on the technical implementation of Snowman. The first sections cover programming languages, development tools and frameworks used in this project. Some sections within this chapter require a general knowledge of the key concepts of Snort described in chapter 4.

6.1 Programming languages

Snowman is primarily written in Python 2.7 [19]. The project group landed on this decision after discussing the aptness of C, C++, Java and Python to this project, with regard to development and processing efficiency.

Due to the limited time-frame of this project, it was vital to get right into the development without having to learn a new programming language or unfamiliar programming concepts. It was also important to find a language which produces high performing programs, since this project aims to be more effective than similar software (see chapter ..). The C and C++ languages were considered because of their performance. The project group also has some theoretical and practical experience with these languages, as they are taught in some courses at Gjøvik University College. These languages were abandoned as the team concluded that they would induce more work because of their complexity, and thereby be too time-consuming.

The final decision fell on Python (why?). no need for the programmer to allocate or free memory. offers a wide range of builtin tools.

A concern was raised that Python, being a high-level language, would maybe not be efficient enough. can write and import C-modules.

A major reason for using Python was Django (see section 6.3) which is a Python framework that greatly simplifies database transactions and generation of web pages.

The most current version of Python available at the start of this project was version 3.3. The project group selected version 2.7, as a break in compatibility [20] between 2.X and 3.X versions of Python cause some problems. According to a recent survey [21] Python 2.X still holds a dominant position among developers, five years after the release of Python 3.0. According to [20], version 3.3 has slightly worse library support. The continuing popularity of 2.X versions implies that many packages are still being developed and maintained for these versions, and these versions only. One example of this is the aforementioned Django framework which is heavily used by Snowman. Development using Python 3.X with Django is possible with the use of a compatibility layer [22], but the project group considered this as a bit cumbersome. Taking into account that the majority of the group was already comfortable with Python 2.X, this came out as the most time-saving choice. The biggest drawback of this decision is that it does not make Snowman future-proof, and it will probably have to be ported to Python 3.X in a long-term perspective.

As for our web technologies, we've utilized HTML and CSS for the design and Javascript for client side scripting.

6.1.1 Standards and guidelines

Python imposes a strict on indentation, as this is actively used in the syntax to delimit blocks of code. The project group has followed standard Python guidelines for best practice programming, with principles as DRY[1] and EAFP[2] (the latter actually caused some unforeseen performance issued described in chapter/section).

Documentation has been a high priority. The project group has aimed to make the code self-documenting by utilizing sensible names on variables, functions and classes to enhance the reader's understanding. As a general rule, are written in camelcase[3] With a high focus on development, the project group has taken advantage of Python Docstrings [23] (documentation strings) which are incorporated into the code, and can therefore be written at the time of coding without taking the focus away from the programming. The Docstrings describe how the code is used, and form the basis of Snowman's documentation. Listing 6.1 demonstrates the EAFP-principle, also note the use of a Docstring at the beginning of the method.

The Python manual encourages the use of exceptions missingobjecterror badformaterror abnormalrule

In Python, methods can communicate with each other using return values

Writing object-oriented code has not been a goal in this project (examples of encapsulation, polymorphism or inheritance?).

```
1  def getSecondMessage(messages):
2    """Returns the second message from messages or
3    empty string if second message does not exist."""
4
5    # Below code demonstrates the principle of
6    # EAFP (1) versus LBYL (2); The execution
7    # of block 1 is faster than block 2 when
8    # message[1] exists. Block 1 is therefore
9    # preferred whenever the try—block is
10   # expected to succeed without an exception.
11
12   # Block 1, EAFP:
13   try:
14     message = messages[1]
15   except IndexError:
16     message = ''
17
18   # Block 2, LBYL (look before you leap):
19   if len(messages) >= 2:
20     message = messages[1]
21   else:
22     message = ''
23
24   return message
```

Listing 6.1: The EAFP-principle.

Coding styles, use of exceptions etc.

[1] "Don't repeat yourself". The principle that code should not be duplicated in a project.

[2] "Easier to ask for forgiveness than permission".

[3] A style convention where multiple words are written as one, with the first letter of each word capitalized, e.g. getAllRules.

6.2 Development environments

Short description of development environments and tools.

6.2.1 Environments

Linux platform. (any cross-platform issues?).

Eclipse platform. PyDev with debugger. PyUnit. TeXlipse and PDF4Eclipse for writing and generating this report.

add-ons written for Eclipse, python, latex which has been used to write this report.

python interpreter has been very useful during the development

Add: Web-browsers, web-debugging etc. Other tools, software used.

6.2.2 Version control

This project has used Subversion (SVN) provided by the IT department at Gjøvik University College for . With SVN, files are stored in a central repository and also locally .

All project files including source code, design documents, meeting minutes and the report documents.

6.3 Frameworks

6.3.1 Django

To facilitate database transactions and the creation of the Web interface, this project uses the Django[6] framework. Django is a Web framework which takes care of database modelling and transactions, and user authentication.

6.3.2 SQLAlchemy

Used on the client side database abstraction...

6.3.3 jQuery

When it comes to using Javascript for client side scripting, jQuery[24] is the way to go. jQuery provides a framework for common javascript operations on the DOM and abstracts away a lot of browser spesific code to give you a nice and clean interface. It also makes AJAX class a lot easier, requireing only code such as

```
\$.get('/web/rules/');
```

instead of

```
function loadXMLDoc() {
    var xmlhttp;

    if (window.XMLHttpRequest) {
        // code for IE7+, Firefox, Chrome, Opera, Safari
        xmlhttp = new XMLHttpRequest();
    } else {
        // code for IE6, IE5
        xmlhttp = new ActiveXObject("Microsoft.XMLHTTP");
    }

    xmlhttp.onreadystatechange = function() {
        if (xmlhttp.readyState == 4 && xmlhttp.status == 200) {
            document.getElementById("myDiv").innerHTML = xmlhttp.responseText;
```

```
15          }
16      }
17
18      xmlhttp.open("GET", "/web/rules/", true);
19      xmlhttp.send();
20  }
```

6.3.4 Bootstrap

Since noone in the group can call themselves a web designer, we decided to take a shortcut on that regard and utilize Bootstrap[25], which is a front-end framework derived from Twitter. It provides us with a lot of ready made CSS and design elements that makes a web page responsive and typographically readable.

6.4 Server architecture

6.4.1 Modules

6.4.2 Database

6.5 Core module

This section will focus on the classes and methods in the `core` module. This module contains the most vital classes of the Snowman server including **`Rule`** and **`RuleRevision`** which hold the data of each rule in the system. Simple classes in this module such as **`Generator`**, **`RuleReference`** and **`RuleClass`**, which are connected to the **`Rule`** class are only briefly explained in this section. For an overview of all objects and functions in this module, refer to class diagram (ref) or appendix (ref).

6.5.1 Rules and rule revisions

Rules in Snowman are split into two parts; the rule and the rule revision. As threats change, or errors in the rules are discovered, a rule may become obsolete and a new revision of the rule is released. The rulestring, message and detection filter might change from one revision to the other, and are stored in the rule revision object along with the revision number. A revision is always connected to a rule object which contains data that is constant for all its revisions, such as signature ID and rule set. The reason behind keeping rules and revisions in two separate objects is to avoid writing the same constant rule data for all revisions. Users of Snowman have the possibility to control how many revisions to store per rule, and whether incoming revisions should be active or inactive by default. This feature allows users to control automatically downloaded rules before they are put into the detection system. Users can also manually activate or deactivate rule revisions, and Snowman will distribute only the newest active revision of a rule.

6.5.2 Rule sets

As stated in chapter 4, the categorization of rules in rule sets is a de facto way of organizing rules. Rules in the community rule set[4] have a “ruleset” option within the “metadata” option in the rulestring which specifies the rule set. In other releases, rules are organized into files where the file name denote their rule set. These two ways of specifying the rule set are both supported by Snowman.

[4] “The Community Ruleset is a GPLv2 VRT certified ruleset that is distributed free of charge without any VRT License restrictions” - http://www.snort.org/snort-rules/

6.5.3 Sensors

6.5.4 Comments

The `core` module contains a comment object which can store comments from users who wish to document their operations, e.g. the user can attach a comment when adding a filter to a rule to describe why the filter was added and which effects it will have on the system. Snowman also generate comments for important events, such as ...

6.6 Update module

The purpose of the `update` module is to fetch and read rules and configuration files from external sources, and populate the database with this information. Effectiveness in updates is a key issue, and a number of deliberate optimizations for performing updates, with the goal of saving time by omitting unnecessary operations, are explained in the subsequent sections. To avoid confusion with the module, the term *update-package* is specifically used to refer to update data coming from external sources.

6.6.1 Sources

A *source* is a supplier of update-packages. Snowman lets users define sources which are either external (i.e. online) or local, meaning that the update-package will either be downloaded from the Web or read from the local system. All information about each source is contained in the `Source` class. The sources determine where the update-packages are located, and an update-package must always come from a specific source.

External sources contain a URL pointing to the location of the update-package, and alternatively also a URL to the MD5-hash of the package. External sources can be configured to pull updates at regular intervals, contrary to local sources which must always be updated manually.

A source can be configured to update at regular intervals, or the user can run an update manually on a source. To make the update process more effective, `update` makes use of an MD5 URL which points to a file containing the MD5 hash of the update-package. This hash is stored for every source (if available) and is compared to any pre-existing hash to verify if the system is up to date or not. `update` will only download and process an update-package when it encounters a new hash.

6.6.2 Files

An update-package may contain various files depending on the source it comes from. This section will describe the files supported by Snowman.

Rule files

A rule file is any file carrying the *.rules* file extension. These files contain raw rulestrings to be processed by Snort. A rulestring is basically a semicolon-separated list of rule options. Specific rule options used in this system are listed in table 13. A badly formatted rulestring given to Snort will cause a crash, and Snort must be restarted.

Classification, Generator and Reference Configuration files

The classification files are identified by their name *classification.config* and contain rule classification strings that are used by Snort to categorize event data. A classification file will normally appear once per update-package, and rarely changes between updates[5].

[5] https://www.snort.org/vrt/snort-conf-configurations/

Data field	Description	Presence
sid	Unique identifier for a rule.	required
rev	Rule revision number.	required
msg	Notification text when rule triggers.	required
classtype	Connects the rule with a specific class of attacks.	required
gid	Generator ID to identify which subsystem generates the event where this rule fires.	optional
ruleset	The ruleset this rule belongs to.	optional
priority	Override default severity set by classtype.	optional
reference	Reference to external resource with information about this attack.	optional

Note: The *presence* column indicates whether a field is required or not by the parser in this system. Information is gathered from the Snort Users Manual [12](ch 3.4).

Table 13: Rule options used in this system.

Sid message files

6.6.3 Running an update

An update is either triggered by the user, or automatically on regular intervals, for a given source. Regardless of how the update process is invoked, it follows the same procedure. As mentioned above, sources will often provide an MD5-hash of the update. When this is the case, the hash is stored in the database. Every new update will compare its own hash to hashes stored in the database, and if an identical hash is found this means that the update has been run before and the update process is terminated. When the hash does not pre-exist in the database, or for sources which does not use hashing, the update-package is downloaded to a temporary directory.

Snowman accepts update-packages in the form of tar-archives[6] containing rule and configuration files, and also supports plain text files containing rules or known configuration entities[7]. Tar-archives are extracted, and folders are traversed recursively

each file in each folder is parsed recursively. For added efficinecy, Snowman creates a hash of all files which is stored in the database. This is later used to see whether any files in a new update has not changed.

In most cases the update-package will contain only minor changes in relation to its predecessor, e.g. only one file is different in the latest package, and parsing all files that has not changed would therefore be a waste. To address this issue and save processing time, the name, path and MD5-hash of every file is stored in the database per source. Every time the update module parses a file, it first creates a hash of the file and compares this with the respective file's hash in the database. If the file is present in the database and the two hashes are identical, the file is considered *not new* and is skipped. The following sections will elaborate on the parsing of files based on file type.

skipping files, sequence of files.

[6] Files that contain compressed files.
[7] Generators, filters, references etc.

parse files with regex.

6.6.4 Rule parsing

Some options can be defined both in the rulestring (inline) and in a separate file (standalone).

If any properties classified as *required* for a rule (table 13) are missing...

Non-existing rule sets, rule classes and generators are stored in the database

After the initial parsing of the rulestring, the parser checks if the rule already exists in the database, and if so whether the incoming revision number is higher than the latest revision of the pre-existing rule. The parsing will continue only in the event of a new rule or a newer revision of an existing rule.

rule moved to new ruleset.

Rule sets

A rule always belogs to a ruleset which can be specified in the raw rulestring. If the parser does not find this attribute in the rulestring, the filename is considered to be the ruleset name.

Threshold

A rule threshold defines a limit and/or threshold for the triggering of a specific rule, and can exist as an inline or standalone option. Rule thresholds are deprecated, and support for this functionality will cease in a future version of Snort, where it is replaced by *detection filters* and *event filters* (see below) [12]. To maintain compatibility with current rules, Snowman supports reading inline and standalone threshold definitions.

Detection filters

Please explain..

Event filters

Please explain..

Suppression

Always standalone.

6.6.5 Classification, Generator and Reference Type parsing

These files contain a sequence of their respective entities (section 6.6.2), and are all parsed in the same manner. If a given entity is not already present in the database, it is created and stored. If the database already contains the entity, the existing entry in the database is overwritten.

Rule Message files

6.6.6 Logging

in Snowman this is translated to one filter of each type per rule per sensor.

If a filter occurs inline, this filter is regarded as a default filter and will be active on every sensor where the rule operates. A user can choose to override the default filter on a sensor by defining a custom filter to be used on the respective sensor.

To stop Snortfrom adding duplicate filters, all inline filters are removed from the rulestrings. Any filters extracted from the rulestring are written as plain text (i.e. as they appear in the original rulestring) in the *filters* field of the rule revision object. This is to ensure that all the original rule options are stored meaning that no data is lost or

discarded.

Filters are removed from the rulestring and transformed into event filter and detection filter objects.

6.7 Client architecture

6.7.1 Modules

6.7.2 Database

6.8 Graphical User Interface

The Graphical User Interface (GUI) is a template- and AJAX-based[8] web application which uses Django's template system to implement the View part of the MVC-principle**??**. Each webpage is a template populated through Python-code before it is delivered to the Web server and ultimately the requesting Web browser.

Additional data that needs to be sent to a Web page, e.g. the user interacts with the GUI, is requested through JavaScript[9] and AJAX-calls and then gets added to the already loaded DOM[10] in the web browser.

6.8.1 Structure

The pages of the GUI is organized in the following structure, which is very similar to the way BHTB organized it's front end.

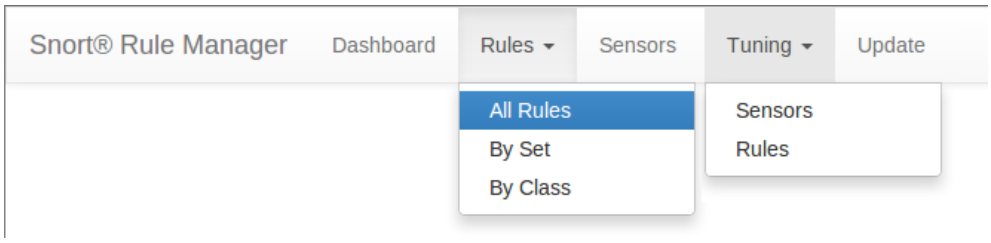


Figure 13: GUI Structure

6.8.2 Design

The GUI is designed with simplicity in mind, which is achieved by using a front-end framework called Bootstrap[25]. This framework provides easy-to-use and ready-made design elements for HTML and CSS[11], which has been crucial for saving time and energy in the project. Consider the example job of creating a <table> element and spending hours setting up and tweaking the CSS for the said table element, we could just add <table class='**table table-bordered**'> and get the table formatted nicely with colors and fonts that are typographically matching.

6.8.3 Overview of central GUI features

Content Prefetching

One of the main features of our GUI design is our usage of prefetching to speed up loading. Instead of loading absolutely everything into the web browsers DOM, like our predecessor BHTB did, we instead initially only load the first page or page elements. This

[8] AJAX is web development technique that allows you to send asynchronous requests to a web server.
[9] A programming language commonly used for client-side scripts.
[10] The webpage content or elements.
[11] A styling language for webpages.

significantly reduces loading times.

We then silently in the background load in any element the user might want to see next, through AJAX calls. When the user then interacts with the GUI, the next elements are shown to the user. The next-next elements are also loaded through AJAX calls so that they are ready.

This ensures that the user always has immidiate response from any interaction done with the GUI, while keeping loading times to a bare minimum.

Rules page

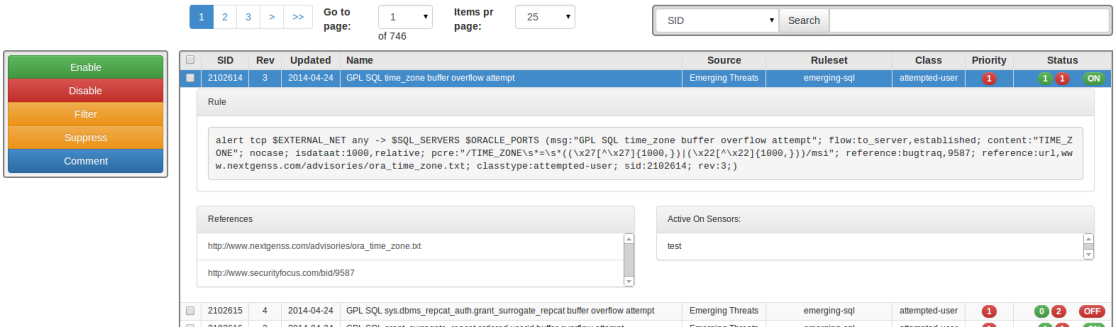


Figure 14: Rules page

This page serves as the complete list of all rules currently residing in the central database. Because the central database can literally contain hundreds of thousands of rules, this list is paginated. Pages are loaded using AJAX calls to prefetch them as needed.

Each row in the list can be clicked on, which expands an area containing additional information about the rule in question.

We anticipate that the main use of this page is to utilize the search function to find a specific rule or rules and then either retrieve information about them or otherwise manipulate them.

Rules can be tuned by checking the checkbox associated with the rule in question and then pressing the button that represent the desired function in the box to the left.

Rulesets page

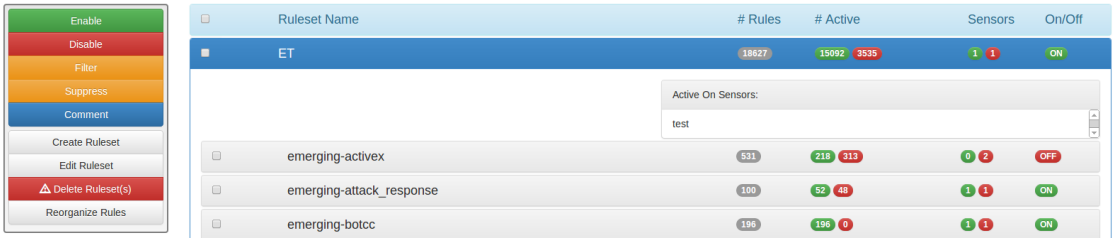


Figure 15: Rulesets page

The rulesets page consists of a list of all the rulesets stored in the central database. Each row contains meta information about the ruleset, such as the number of rules in the ruleset and so on.

A special feature of the ruleset page is that since Snowman allows you to structure rulesets in an hierarcical tree, the ruleset list will initially only show the tree roots. If the ruleset has child rulesets associated with it, those will be listed once you click on the ruleset and open up the additional information area.

The list controls also contain functionality for creating, editing, deleting and reorganizing the rulesets, as well as the ability to turn them on or off either globally or on a per sensor basis.

We also have a very similar page for the sensor list, which also allows you to arrange sensors in a hierarchical tree. Because it is virtually the same, it is not detailed in the report.

Updates and changes

One of the big requests from the client when it came to improvements of features of BHTB was the need to have more feedback over the update process, as BHTB only told you about the status after it was done, leaving the user in an information-limbo.

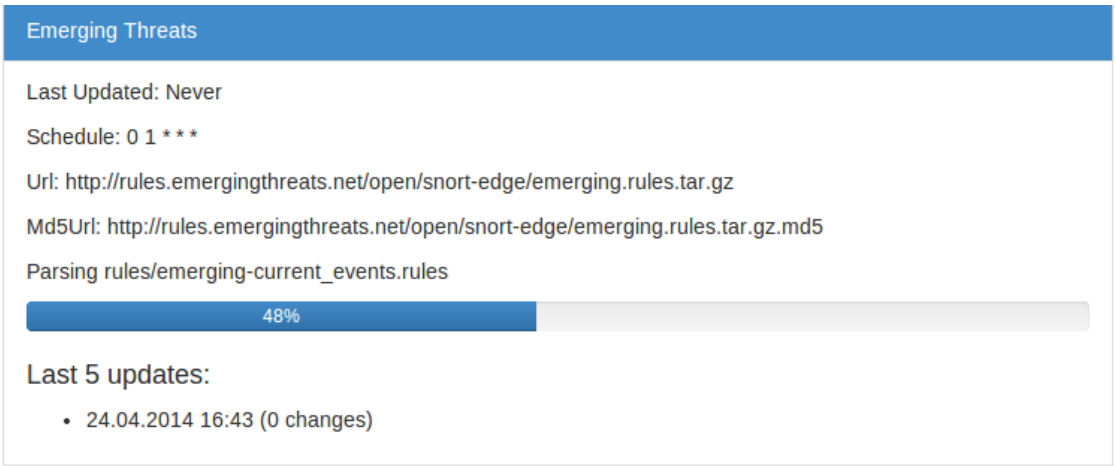


Figure 16: Update progressbar

To solve this, we added a continually updating progressbar which tells the user what the system is currently doing and giving a rough estimate of the percentage of completion.

Another requested feature was the ability to see what was changed during an update. We provide a list for this which contains each update and it's various changes.

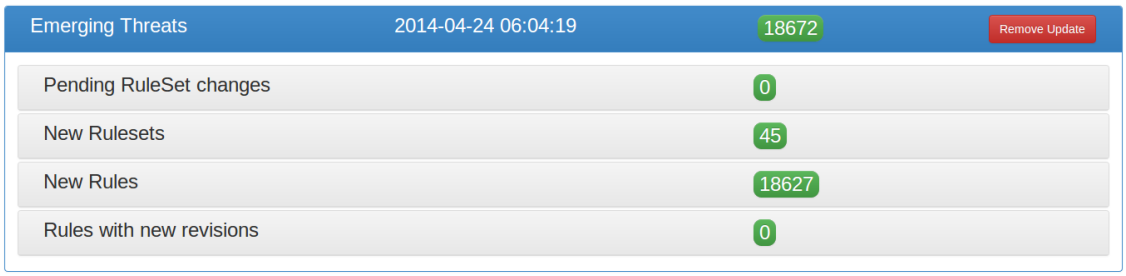


Figure 17: Update changes

The list details any new rulesets, rules or rule revision an update might have brough with it, as well as a list of rules that have been moved from one ruleset to another. Once the user has finished checking out the changes from one update, the user can remove the update from the list. This makes the list a backlog tool, rather than a perpetually long list or some sort of time limited list (say, updates in the last 7 days or something similar). This gives the user a good overview of what has changed since the last time the user checked the list.

Form validation and feedback

One challenge that always face web frontends are forms, as they are a means to alter data in the database and thus become a security challenge. So when you utilize forms, it is important to protect the database against both malicious attackers and user errors.

There are three basic steps to implement protection against these security issues, which is to (1) validate the data provided by the form, (2) make sure the user can't input something it shouldn't and (3) to never let any raw strings or values from the form be directly inserted into a database query.

The third protection is perhaps the most important as it leaves you vulnerable to SQL injection attacks, which are attacks where SQL code is input into the form, letting the attacker carry out his/her own queries, leading to total database compromise. This problem is solved through what is called prepared statements, which makes sure a query doesn't do anything it's not supposed to before the query is actually carried out on the database. This protection is implemented in our system through Django's internal protection mechanisms.

The first and second protection is something we have implemented ourselves at both the frontend and the backend. We make sure that the user can never input invalid values into any form by having Javascript validate the input. So if a user inputs characters where only numbers are allowed, the user will be notified immediately and not allowed to submit the form until it is corrected. And this is done before anything is ever send from the browser to the server.

But if the user has input all valid values, we still validate all the data at the backend

before we store anything in the database. This means checking that any referenced object ID actually exists or making sure the provided IP address is actually valid (within the 0.0.0.0/0 - 255.255.255.255/32 range). If anything is out of place, a message is sent back to the user about it.

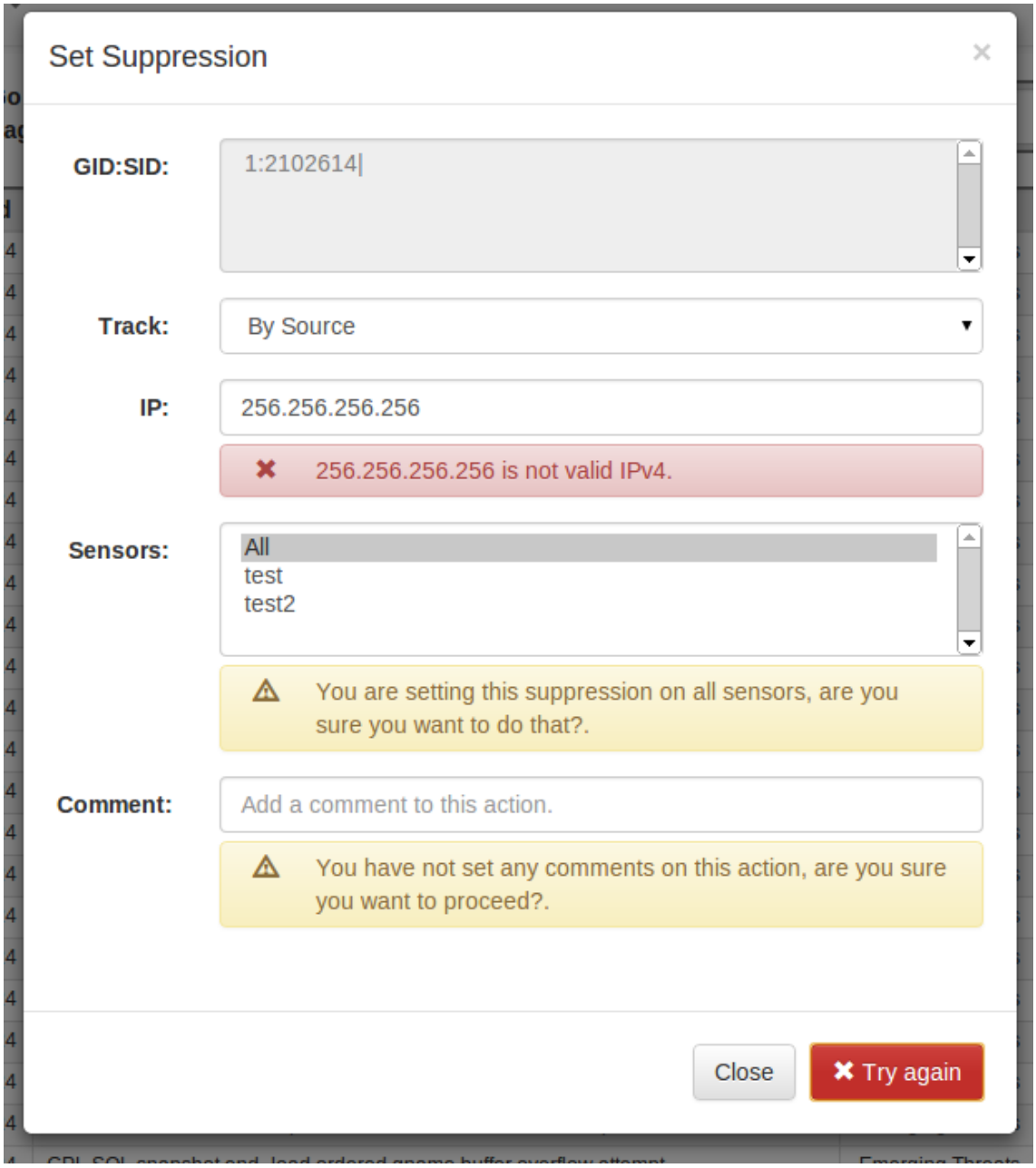


Figure 18: Form validation

6.9 Logging

Describe how logging is implemented in the software.

6.10 Configuration file

During the design of Snowman the solution to some issues was not immediately clear. As an example, consider the automatic update of new rule revisions. Some users would like to have the new revisions put into production, but others might want to keep the new revisions inactive until they are considered good and ready.

It was decided that these types of issues should be specified in a configuration file.

For these types of issues, where the project group remained inconclusive

not clear whether it would be best to do it

(e.g. should new rules always assume a deactivated state or the opposite). These types of issues

Describe the purpose and structure of the config file.

Some update-packages may contain source code files or modules which are of no interest for Snowman.

Users can choose between using a whitelist (parse only whitelisted files) or a blacklist (parse all but the blacklisted files)

6.11 Documentation

How the software is documented.

6.12 Deployment

Describe installation/packaging.

7 SOFTWARE TESTING

Her beskrives hvordan opplegget rundt testing av applikasjonen har vært. Dette omfatter planlegging, valg av teststrategier, gjennomføring av tester og beskrivelser av sentrale feil funnet under testing, samt måten feil ble rettet opp på.

7.1 Strategy

The overall strategy of the project when it comes to testing has, first and foremost, been to have regular meetings from the client where we demonstrate live demo's of the software. This has lead us to either confirm things as feature complete or created a list of things to either fix or improve, along the way.

We have also not had any specific regimes when it comes to bugfixing, we have mostly just handled them when they have been discovered them.

You could describe our software testing strategy as ad-hoc based, as is quite common in agile projects. Our main focus has been on the performance testing.

Unit testing has not been used extensively in this project since the development team has limited knowledge and experience with this type of testing. It has been used, however, to test that data from an update is correctly inserted into the database. Source code listings of all unit tests in this project can be found in appendix.

7.2 Description of Problems

7.2.1 Rule insertion

Snowman extracts various types of data from the rulestring and inserts this into the database in different tables. Since an update can contain thousands of rules it proved very difficult to effectively verify that all rules were correctly inserted into the database. To address this, a test suite was set up executing a dummy update with an example rulestring containing all possible options.

```
def test_updateRule(self):
  # Insert the rule
  self.update.updateRule(self.rulestring, "example.rules")

  try:
    # Verify that all related objects exist
    rule = Rule.objects.get(SID=1, active=True, priority=10)
    generator = rule.generator
    ...
  except Rule.DoesNotExist:
    self.fail("Rule does not exist")
  except Generator.DoesNotExist:
    self.fail("Generator does not exist")
  except ...

  self.assertTrue(rule.active==True)
  self.assertTrue(int(rule.priority)==10)
```

```
18    self.assertTrue(generator.GID==1)
```

These tests were also very useful during the refactoring of the Update module.

7.2.2 List Pagination

One problem that has plagued the software throughout the development has been the pagination functionality in the GUI. Due to it using AJAX to fetch other pages in the background, if you changed pages so fast that the system couldn't catch up, it would just end up never loading anything due to AJAX's asyncronous nature[1]. This has been remedied by using some logic to check for outstanding AJAX calls but a better solution would be to implement some other mechanism or framework for pagination. The remedy has also put some extra latency on the rule list as waiting for the AJAX calls might take some time.

7.2.3 The Ever Changing Number of Changes

One smaller bug that popped up was on the Changes page where the number of new rulesets, rules and rule revisions there has been in the latest updates were displayed. Initially, only the rule revision was tied to the Update object and we then traversed from the revision object to the rule object and count how many revisions that rule had, in order to determine if the rule was new[2]. But using this method led to a undesired outcome when a "new" rule gets a new revision in later sequent update. As the count for revisions is done real time, this rule will suddenly not be listed as new anymore in the old update.

This problem was solved by tying all relevant new objects to the Update object and then just list those relations. The downside of this is of course a lot more relations to store in the database, but the Changes view was one of the more important features so a camel had to be swallowed.

[1] The code does not wait for AJAX to finish before moving on.
[2] A shiny new rule will always only have one revision.

8 PERFORMANCE ANALYSIS

8.1 Theoretical analysis

The purpose of the theoretical performance analysis is to examine the most important features of Snowman and similar software, to form opinions on the expected performance of each software. The teoretical analysis is used to into concrete benchmarking tests presented in section 8.2.

BHTB, Snortmanager and two versions of Snowman are analysed.

8.1.1 Effectiveness of update processing

As described in chapter (impl?) Snowman considers during the rulestring parsing. BHTB writes rules to one table, whereas a rule object in Snowman has multiple relations to other objects and thus several database tables are written to during an update. It is therefore expected that BHTB and Snortmanager exhibit a more effective update processing.

File-to-rule ratio

A sample update package[1] contained on average [number of rules per file] rules per file. Snowman is designed to

the only extra overhead for Snowman is to open each file. It is therefore expected that the difference in processing time for Snowman will not exceed 1 percent.

Incremental updates

One of the expected strengths of Snowman lies in processing incremental updates, i.e. updates containing only a small number of changes. It is significant to gauge performance for incremenal updates, as most real-life updates usually contain between 10 and 100 changes[2].

Storing the checksum for every individual file[3] is unique for Snowman. It allows Snowman to completely ignore files without changes, which means that only the files with rule changes are parsed. This is particularly useful as the current official Snort rules contain [number of files] files[4]

(how does bhtb and snortmanager cache?) caching of files

8.1.2 Effectiveness of rule distribution

8.1.3 Linearity in performance

8.2 Practical analysis

Since Snowman is similar to BHTB and Snortmanager, it has been important to benchmark their performance in various formal test cases in order to highlight the strength and weaknesses in each software solution. The purpose of the tests is to reach final conclusions on key questions regarding the performance of Snowman, and to point out where the difference between the three software packages lies.

[1] link to update package used in testing

[2] Ten updates from 20.03.2014 to 17.04.2014 contained on average 36.8 new and 53.9 updated rules. *Source: http://www.snort.org/vrt/advisories*

[3] Checksum is stored only files that are parsed.

[4] link to rules.

To get a picture of how the software performs on different systems, it was tested on three architectures (table 14). To minimize error, each test was run five times where the mean value of the runs is considered the valid test result. All tests can be reproduced with the test procedure documentation in appendix.

8.2.1 Definitions

The testing is split up into *test suites*, which contain a *problem* in the form of a question. The problem forms a clear description of what is to be tested, and how results should be evaluated. A test suite also contains one or more *test cases* which exist to address the problem. Test cases represent individual tests, and describe the input data and constraints. Test cases are carried out on the different systems, and results from each test case are reffered to as the *test results*.

8.2.2 Environment

All tests are carried out on three systems, named *S1*, *S2* and *S3* (table 14). These systems differ in processing power, available memory and storage technology, and the results will indicate what impact these factors have for software performance. The performance testing in this report is also prone to error, as tests might get disturbed by other background processes etc. Testing on three systems gives results with a higher confidence, as results are checked aginst each other for inconsistencies. When an inconsistency in results is found, the test is run again to see if the inconsistency prevails.

System	Processor type	Memory	Storage
S1	Intel® Core™ 2 Solo, 1.4GHz	4GB RAM	5K RPM HDD
S2	Intel® Core™ i7, 6x3.2GHz	12GB RAM	10K RPM HDD
S3	Intel® Core™ i5, 4x3.2GHz	16GB RAM	SSD

Note: See appendix for full details.

Table 14: Test systems.

8.2.3 Test suites

Even though Snowman, BHTB and Snortmanager share the same goal of being a rule management system, it is not immediately obvious how they can be tested against each other to compare their performance. The first step in the performance testing has been to identify the areas where the three software packages share, to a certain extent[5], the same functionality. In the test planning, considerations have been made to ensure that the tests hold an acceptable degree of fairness (i.e. tests are designed so that none of the software enjoys any improper advantages or disatvantages), and that the tests are pertinent to the daily operation of the software and the Snowman requirement specification.

Snowman, BHTB and Snortmanager all have the functionality to download and process an update from an external source. The first group of test suites presents different test cases to benchmark how each software performs in update processing.

[5] Snowman, BHTB and Snortmanager have been built to tackle many of the same challenges, however, the actual implementations may differ severly. For an extended walk-through of differences in implementation, refer to chapter 2, and chapter 10.

8.2.4 Update processing tests

The purpose of the tests in this section is first and foremost to test how quickly the software can process an update, and if the processing time is linear to the update size. It is interesting to test for linearity as this may help point out programming errors. To get a complete picture of the performance, the tests are set up with different constraints such as update size, number of files and number of changes from one update to the next.

Tests numbered 1 through 4 use an artificially generated update package where all rules contain the same data[6].

Test case	In test	Description
SYN-1	Test 1, Test 2	Blank database.
SYN-10		
SYN-100		
SYN-1000		
SYN-10000		
SYN-1000-B	Test 3	All rules in one file.
SYN-1000-C	Test 4	Pre-populated database. No changes in update.
SYN-1000-D	Test 5	Pre-populated database. Update has one rule change.
SF-28064	Test 6	Blank database.
SF-28064-B	Test 7	Pre-populated database. Update has one rule change.

Note: Rules are split into files with ten rules per file unless other is specified.

Table 15: Test cases in update processing.

Test 1

Problem: How fast can the software process an update?

The purpose of this question is to determine the average processing time per rule. Results are ranked from short to long processing time, where shorter is better.

Test cases:

To address this problem, five test cases were introduced. The tests contained an artificially generated update with 1, 10, 100, 1000 and 10000 rules, respectively. Rules were split into files with ten each (table **??**). Each test case represents one update, and the names denote the number of rules, files and rules per file in the following structure: *rules-files-rules per file*.

Test 2

Problem: How strong is the linear correlation between processing time and the size of an update?

This question rises from the requirements specification of Snowman (chapter 3), where it is stated that Snowman shall exhibit a linear increase in processing time in correlation to the number of rules in the update (where the number of rules depicts the update size).

[6] Only signature IDs are different. The rules contain a message and a classtype, see appendix for full description.

Results are ranked from strong correlation to no correlation, where stronger is better.

Test cases:

This test uses the same test cases as Test 1.

Test 3

Problem: Does raising the ratio between files and rules increase processing time?

Updates can come with the rules either gathered in one file, or distributed over several files. This question is asked to examine the difference in processing time for these two scenarios. Results are ranked from no difference to big difference, where less difference is better.

Test cases:

This test compares the *1000-100-10* test case from Test 1 with a test case with the same number of rules in *one* file (1000-1-1).

Test case	Description
SYN-1000-B	All rules in one file.

Table 16: Test cases in Test 3.

Test 4

Problem: Is the processing time shorter when the update contains no new rules?

This is a test for update processing when the update is not new (i.e. it has been completely processed by the system before). To minimize system load, it is essential that already processed updates are (in practice, they should be neglected) by the system. Results are ranked from shortest to longest processing time, where shorter is better.

Test cases:

This is a compound test where a test case is run two times in succession. The 1000-100-10 test case from Test 1 is run first to populate the database, before it is run again.

Test 5

Problem: Are the processing times for two updates identical when only one rule has changed between the updates?

This question seeks to determine how the software handles an update when there is only one change from the previous update. This is an important factor to consider, since an update usually contains only a small amount of changes from previous updates. Results are ranked from short processing time for second update to no difference in processing time, where shorter time is better.

Test cases:

This test suite consists of two test cases, one with an artificially generated update and one with an update from a real source. The test cases 1000-100-10 and from Test 1 are used. Each update is run twice; the first run populates the database, and the second run contains one change from the previous run (one of the rules has an updated revision number).

8.2.5 Rule distribution tests

8.3 Optimization

Include here the changes made between Snowman v0.4 and 0.5.

```
from core.models import RuleClass

def updateClassification(type, description, priority):
  try:
    # Update existing classification
    ruleclass = RuleClass.objects.get(classtype=type)
    ruleclass.description = description
    ruleclass.priority = priority
    ruleclass.save()
  except RuleClass.DoesNotExist:
    # Add new classification
    ruleclass = RuleClass.objects.create(classtype=type, ...)
```

Listing 8.1: Example code

9 RESULTS

This chapter presents the results from the performance testing.

Mean times calculated from five sequential test results. See appendix for a complete listing of results.

Test case	System	Snowman v 0.4	Snortmanager	BHTB
SYN-1	S1	1058.179	228.617	125.236
	S2	622.640	93.403	65.281
	S3	151.946	36.960	18.175
SYN-10	S1	2289.518	265.275	148.323
	S2	1668.142	107.554	72.025
	S3	327.655	37.569	19.711
SYN-100	S1	17519.385	1251.275	185.970
	S2	12197.192	139.005	84.913
	S3	2356.610	159.113	25.024
SYN-1000	S1	167645.267	9470.460	568.333
	S2	114617.561	460.394	158.351
	S3	22830.358	1360.276	60.313
SYN-10000	S1	1795175.128	92428.555	3606.043
	S2	1162468.835	5105.158	910.172
	S3	228038.066	13523.887	440.334
SYN-1000-B	S1	147170.285	1836.396	445.476
	S2	113901.478	496.204	157.817
	S3	20367.838	228.044	49.908
SYN-1000-C	S1	296.272	1763.480	7.816
	S2	186.215	829.230	4.543
	S3	56.152	205.956	1.310
SYN-1000-D	S1	1371.722	64100.000	563.682
	S2	9682.763	31999.710	85.289
	S3	16998.276	10716.689	83.275
SF-28064	S1	9175886.000	43391.700	5638.424
	S2	3817584.070	21127.828	1715.521
	S3	1267483.848	8039.046	1017.561
SF-28064-B	S1	28200.953	239878.897	5606.518
	S2	22673.225	80009.077	2396.780
	S3	5260.553	63626.550	1004.424

Table 17: Test results for update processing.

10 ANALYSIS OF RESULTS

Diskusjon rund resultatene. Det skal gjøres en evaluering av hvorvidtvalgt løsning. Avvik drøftes. det er samsvar mellom kravspesifikasjonen satt opp og

An update package usually contains only a small number of changes from the previous version

11 CONCLUSIONS

De slutninger som kan trekkes fra resultatene og deres diskusjon skal være logisk og nøkternt oppbygd. Etterrettelighet er vesentlig. Konklusjoner skal fremkomme klart og uten tvetydigheter. Eventuell usikkerhet påpekes. Grunnlaget for slik tvil skal redegjøres under kapittel 6a/6b. De viktigste konklusjoner blir presentert i kortfattet versjon under "Sammendrag". Prosjektgruppen bør ta stilling til hva de faglige resultatene kan brukes til, hva gruppen faglig har lært av dette arbeidet, hva som evt. kunne vært gjort annerledes ved en gjentagelse av utviklingsarbeidet, og hvordan eventuelt videre arbeid rundt samme eller beslektet tema bør gjøres. Gruppen bør også kommentere den subjektive opplevelse av hovedprosjektet, og hvordan gruppeprosessen har vært. Kort sagt: Egenevaluering av gruppearbeidet.

Some performance is sacrificed for more functionality.

11.1 Future work

Can use [1] to check rules for errors before using them in production.

11.2 Discussion

11.2.1 Results

- we discovered that performance is hard
- it was nice to abstract the database
- we created something that works, but lacks a few features
- performance goals was not met, but functions are in place
- GUI is hard
- We learned stuff - something something darkside

11.3 Future Development

At the start of the project we were ambitious with our project goals and the client had a long list of functionality they wanted implemented. While a large chunk of the functionality made it to our latest version, we had to sacrifice some due to time constraints.

And that were the functions we put in the "if time permitting" pile from the start, the ability to comment rules, the ability to write new custom rules and to have some sort of way to check for errors in the rules before they are distributed. There is, however, no problems adding these features in future work, as we've strived to make the program modularized.

There are also more optimization work that could be done to speed things up, especially when it comes to fetching lists of rules.

The GUI could probably also benefit from more work, as it has a few quirks and odd bugs. It would probably be beneficial to find some other way of doing pagination, as the implementation in the current iteration is not optimal. There's also some cosmetic improvements to be done to tighten up the looks, there are a few places where theres

[1] http://leonward.wordpress.com/dumbpig

too much clutter once you've navigated and opened too many things.

We also compiled an improvements list from our client on a few matters, which is available in appendix [**?**].

11.4 Group Evaluation

11.4.1 Introduction

At the beginning of this project we all agreed on that we wanted to reach for that often unreachable top grade on this project. To accomplish this, we've worked countless hours to the best of our abilites and we've had virtually no issues or problems to sidetrack us as a group, so we've been able to fully focus on the task at hand.

The group has worked well as both a team and as individuals, drawing from each others strengths and expertise. We've created something that could very well either live on in service of the information security community or inspire others to further improve or recreate.

The work that has been done over the past few months is something the group can be proud of.

11.4.2 Organization

While we had a designated group leader, we never really got in a situation where we needed one. Most of our decisions or problems were solved as a group through discussion and a democratic approach and that worked well for our group.

It seems like we hit the nail pretty much dead on when it came to our project plan, which we have more or less followed throughout the entire project. The development cycle lengths and intermittant meetings has allowed us to be very flexible and agile in our development, as well as being more or less on schedule.

Spending some time planning everything before we started programming gave us a good head start and even saved us some time writing this report as we had a lot of its components already made. When we completed the initial planning phase, we held a "project decision meeting" with the client where we introduced the system models we had made so that the client could decide what they wanted or not. We felt this was a good idea, as we could then get some valuable feedback before we started programming.

Our approach to the development cycles was mostly as described in section 1.9.1, we conducted syncronizational meetings once or twice a week to chart out the course and keep up to date with what the others were doing. But for the most part, because we had devided the software into modules, we worked individually. We feet this was the way to go, since we could all work comfortably at home in front of our favourite computer, being more productive that way.

When it came to the thesis, we set a good month and a half of our project time for it and we think it paid off, especially as we expanded it to include the performance testing part. We also devided the writing part into three draft cycles, so we could get valuable feedback from our mentor and client.

11.4.3 Work distribution

For the most part our work distribution has been a "free for all", meaning the group members picked freely from what they wanted to work on during the development cycle, within the objectives we set for that cycle. This worked out fine for the most part, but we feel we probably could have been slightly more aggressive with how many features we

finished within the cycles. This left with with a bigger workload towards the end of the development as opposed to the beginning.

11.4.4 What could have been done differently

As most bachelor thesis' write towards the end, "if only there was more time". In hindsight, we probably could've conducted our performance testing earlier in the project, so that we could have more time to implement optimizations. We should've also had a more strict approach to what we were supposed to work on for the different development cycles, planning that out earlier as well.

11.4.5 Subjective views

Thomas Nyheim

Skriv om folelser og slikt

Eigil Obrestad

Skriv om folelser og slikt

Eirik Skogstad

Skriv om folelser og slikt

11.5 Conclusion

This was fun and stuff.

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